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Long-Term Mathematical Model Involving Renal Sympathetic Nerve Activity, Arterial Pressure, and Sodium Excretion

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Abstract—This paper presents a physiological long-term model of the cardiovascular system. It integrates the previous models developed by Guyton, Uttamsingh and Coleman. Additionally it introduces mechanisms of direct effects of the renal sympathetic nerve activity (rsna) on tubular sodium reabsorption and renin secretion in accordance with experimental data from literature. The resulting mathematical model constitutes the first long-term model of the cardiovascular system accounting for the effects of rsna on kidney functions in such detail. The objective of developing such a model is to observe the consequences of long-term rsna increase and impairment of rsna inhibition under volume loading. This model provides an understanding of the rsna-related mechanisms, which cause mean arterial pressure increase in hypertension and total sodium amount increase (sodium retention) in congestive heart failure, nephrotic syndrome and cirrhosis.

Keywords—Long-term cardiovascular system model, Hypertension, Congestive heart failure, Nephrotic syndrome, Cirrhosis.

INTRODUCTION

It is known that kidneys play an important role in the maintenance of the long-term arterial pressure and body fluid volume regulation. During the first half of the 20th century it was accepted that renal nerves have little functional importance. However, later, it has been demonstrated that renal nerves innervate on functionally important parts of the kidney. Next it was shown that renal nerves directly affect some important kidney functions.[15] Furthermore, clinical and experimental observations indicate an increase in renal sympathetic nerve activity (rsna) in cases of congestive heart failure, nephrotic syndrome and hepatic cirrhosis, which have common clinical presentation of sodium retaining edema formation. The contribution of increased rsna to renal sodium retention in diseases mentioned above was demonstrated by several clinical and experimental studies.[12]

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In view of the clinical importance of the long-term interaction between rsna, arterial pressure and sodium excretion, a model has been developed, which integrates and improves the mathematical models of the cardiovascular system existing in the literature.

The most extensive mathematical model of the cardiovascular system known in the literature is the one developed by Guyton *et al*.[22] Other models related to the cardiovascular system include Uttamsingh's and Coleman's models[7,40] which basically focus on the renal system. Although these models include most of the mechanisms responsible for blood volume control, the effects of the renal nerve are accounted for in only Guyton's model[22] and this in a very limited manner. Uttamsingh's and Coleman's models provide a more detailed representation of the kidney, whereas Guyton represents it in a more compact manner.

The model proposed in this paper integrates the advantageous parts of these models and also introduces the renal sympathetic nerve effect in a more extensive manner. The organization of the paper is as follows: Section II gives a comparison of the mathematical models of the cardiovascular/renal system known in the literature. In section III the proposed model is presented. Section IV includes the simulation results of the proposed model and their comparison with clinical and experimental data. Section V the conclusions are summarized including the limitations of the model.

SOME MATHEMATICAL MODELS OF THE CARDIOVASCULAR/RENAL SYSTEM

In 1967 Guyton published his first cardiovascular model[20] where he provided a basic understanding of the mutual effects of basic cardiovascular variables like mean arterial pressure, total peripheral resistance, cardiac output etc. His later model[22] can be considered as the first extensive mathematical model of the long-term (a few weeks) behavior of the cardiovascular system simulated on the computer. This model consists of a large set of coupled non-linear differential equations. The representation as a block diagram includes, beside some integrators and gains, blocks with outputs representing physiological variables. Such blocks express the physiological variable under consideration as a function of other variables. When obtaining these functions, Guyton has integrated experimental and clinical data from a variety of sources. Since the model is supposed to account for the long-term behavior of the system, Guyton *et al.* have left the fast systolic and diastolic dynamics out of the scope and have represented the arterial pressure in terms of the average of the systolic and diastolic pressures. Similarly, dynamics much slower than observable within the considered time scale (i.e., months) are disregarded, i.e., the corresponding variables are taken as constants.

In 1970's several attempts have been made to model various aspects of the cardiovascular/kidney system.[3,6,27] However, these will not be considered in this paper, since they do not introduce any novelty to Guyton's model as far as the renal sympathetic nerve activity is concerned.

In the subsequent years Uttamsingh has developed a detailed mathematical model of the long-term renal system dynamics[40] also including some basic cardiovascular dynamics. The main novelty in Uttamsingh's model[40] is the dependence of renin secretion on the macula densa sodium flow.

A later and more extensive contribution to renal system models is the one by Coleman[7] which focuses on the sodium flow in the macula densa, and provides a detailed representation of the renin secretion and tubulo-glomerular feedback mechanisms (Fig. 1). It also includes the effect of angiotensin on the tubulo-glomerular feedback signal.

Table 1 shows a qualitative comparison of the causal relations between some variables and renal processes as used by various models.

PROPOSED MODEL

The objective of this study is to develop a computer-based model of the long-term cardiovascular system response to increased renal sympathetic nerve activity and to inhibition of renal sympathetic nerve activity. The model combines major parts of Guyton's extensive cardiovascular model with two renal models in the literature (Coleman's and Uttamsingh's models) and improves them with some further experimental findings from the literature, such that the simulation results sufficiently resemble the behavior of the arterial pressure and sodium excretion under long-term rsna increase and under volume loading/rsna inhibition. Short-term dynamics are not considered in this model.

Guyton's models[20,22] provide a very good and extensive basis for any long-term model of the human cardiovascular system. However, keeping the aim of accounting for the renal sympathetic effects in mind, the model presented in Guyton *et al.*[22] needs to be improved, because it represents the renal tubular system as a single block and is therefore not suitable for analyzing the influence of renal sympathetic nerve activity at different parts of this system. In Guyton's model,[22] the effect of the renal nerve on kidney functions is confined to the influence on the afferent arteriolar resistance (Table 1). However, in DiBona *et al.*,[12] it has been reported that renal nerves directly affect some activity in different parts of the renal tubular system: Renin secretion from the juxtaglomerular complex and tubular sodium reabsorption. The proposed model adopts the representation of the renal tubular system in terms of its anatomical subregions as given by Coleman[7] (Fig. 1), which allows not only a detailed account for different effects of the renal sympathetic nerve activity (rsna) but also a detailed study of sodium reabsorption from different parts of the renal tubular system.

Furthermore, some hormonal effects represented in closed form in Guyton's model have been replaced by more detailed representations in Uttamsingh *et al.*[40]

Thus, based on medical knowledge, the proposed model integrates to Guyton's models for the human cardiovascular system[20,22] the above mentioned physiological functions as represented in Coleman's and Uttamsingh's kidney models and some quantitative findings from medical literature. These quantitative findings include:

- Dependence of right atrial pressure on atrial natriuretic peptide[31]

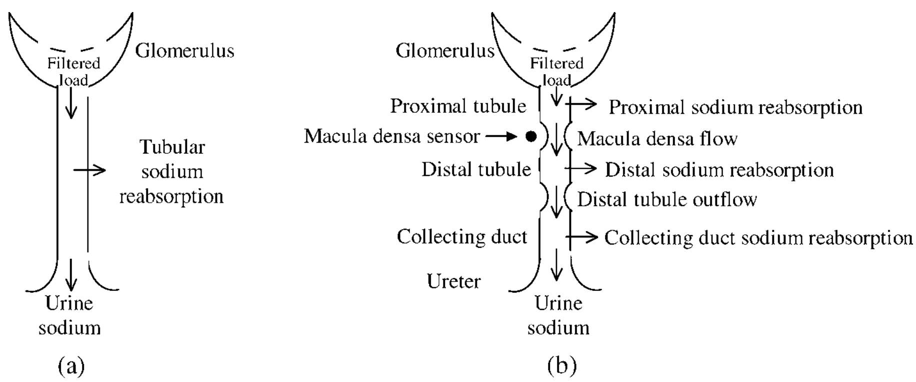


FIGURE 1. Renal tubular system as modeled by (a) Guyton and (b) Coleman.

TABLE 1. Comparison of different models in the literature with respect to some renal mechanisms.

	Guyton's model (Guyton, 1972)	Uttamsingh's model (Uttamsingh, 1985)	Coleman's model (Coleman, 1992)	Proposed model
Factor(s) influencing renin secretion	—	• Macula densa sodium flow	• Macula densa sodium flow	• Macula densa sodium flow nerve • Renal sympathetic activity
Factor(s) influencing angiotensin secretion	• Sodium concentration • Renal blood flow	• Renin concentration	• Renin concentration	• Renin concentration
Factor(s) influencing aldosterone secretion	• Potassium concentration • Sodium concentration • Angiotensin concentration • Arterial pressure	• Angiotensin concentration	• Angiotensin concentration	• Potassium • Sodium concentration • Angiotensin concentration • Arterial pressure
Factor(s) influencing antidiuretic hormone secretion	• Right atrial pressure • Sodium concentration • Autonomic nervous system activity	• Sodium concentration • Excess fluid in extracellular compartment	—	• Right atrial pressure • Sodium concentration • Autonomic nervous system activity
Factor(s) influencing atrial natriuretic peptide secretion	• Natriuretic factor[a]	—	—	• Right atrial pressure
Variable(s) affected by renal sympathetic nerve activity	• Afferent arteriolar resistance	—	—	• Afferent arteriolar resistance • Renin secretion rate • Tubular sodium reabsorption rate

[a] In 1970s although the existence of natriuretic peptide regulation was recognized, [27] the detailed regulation mechanism was not known yet. Therefore, Guyton has used a hypothetical regulation loop, which he called "natriuretic factor."

- Effect of atrial natriuretic peptide on sodium excretion[25]
- Dependence of rsna on right atrial pressure[11] and arterial pressure[10]
- Effects of rnsa on the renin secretion rate[8] and the tubular sodium reabsorption rate[8,35]
- Data on sodium excretion and renin concentration[36,39]
- Since sodium flow in macula densa and renin secretion rate do not lend themselves to practical measurement, some assumptions have been made in order to estimate them from measurable variables reported in the data given in.[36,39] These assumptions are as follows:
  - (i) It has been assumed that the changes in the sodium flow in macula densa is similar to the changes in sodium excretion in urine, with some modulation by the distal tubule and collecting duct sodium reabsorption rate. Anatomically, after macula densa, sodium goes through the distal tubule and the collecting duct [Fig. 1(b)], where some part of it is reabsorbed, and then leaves the body in the urine. This assumption is actually a consequence of a more basic assumption that both the distal tubule sodium reabsorption rate and the collecting duct sodium reabsorption rate are constant percentages of the respective inflows. The assumption on the fractional distal tubule sodium reabsorption rate is justified by the experimental findings of Barba *et al.*,[1] who report an increase of only 3% (small enough to be assumed as 0%) in the fractional distal tubule sodium reabsorption as a response to dropping the sodium intake down to 62.5% of its normal value. On the other hand, the assumption on the fractional collecting duct sodium reabsorption rate is based on the following argumentation of Mejia *et al.*[34] "because of relatively small amount of NaCl reabsorbed in the collecting duct system, control mechanisms at this site can only mediate relatively small changes the NaCl excretion rate."
  - (ii) It has been assumed that at steady state the percentage change in the renin secretion is proportional to renin concentration change in the blood. The justification of this assumption is based on the fact that at steady-state renin secretion and destruction rates are equal. Consequently, a percentage change in steady-state renin secretion can be said to cause the same percentage change in the steady-state renin concentration.
  - (iii) It has been assumed that renin is secreted only from the kidneys according to.[24]

All basic assumptions in the proposed model, including those mentioned above, can be listed as follows:

- The kidneys are modeled as if they constitute a single huge nephron.
- It is assumed that the change in sodium flow in macula densa is similar to the change in sodium excretion in urine, with some modulation by the distal tubule and collecting duct sodium reabsorption rate.

- It is assumed that at steady state the percentage change in the renin secretion is proportional to the percentage change in renin concentration.
- It is assumed that renin is secreted only from the kidneys.
- It is assumed that the direct influence of renal nerves on tubular sodium reabsorption is confined to the proximal tubule.
- It is assumed that there is no baroreceptor mediated rsna adaptation.
- It is assumed that baroreceptor mechanism has no influence on renin secretion.
- The effect of potassium concentration on aldosterone secretion is assumed to be constant.
- The sodium reabsorption effect of aldosterone is assumed to be confined to distal tubule.

Throughout the model the hormone regulation mechanisms have been represented in accordance with Guyton's approach. This approach is based on a simplified model, which relates the hormone concentration to the hormone secretion level. Since at the steady-state the same amount of hormone is destroyed as is secreted, a percentage change in the hormone secretion will result in the same percentage change in the steady-state value of the hormone concentration. This steady-state concentration will obviously be reached with a certain time delay. This behavior can be approximated by an exponential convergence with a specific time constant T, which corresponds to the time delay for the concentration to reach approximately 63% of the steady-state level.

From control theory point of view, this exponential convergence of the percentage change in hormone concentration (x) to the percentage change in hormone secretion level (x_s) can easily be represented by a first order differential equation $\frac{dx}{dt} = \frac{1}{T}(x_s - x)$. Instead of the differential equation, the above relation is used in the integral form in the simulations.

$$x(t) = \frac{1}{T} \int_0^t (x_s - x)d\tau \tag{1}$$

Figure 2 shows the block diagram of the proposed model. Each block represents the dependence of a physical or

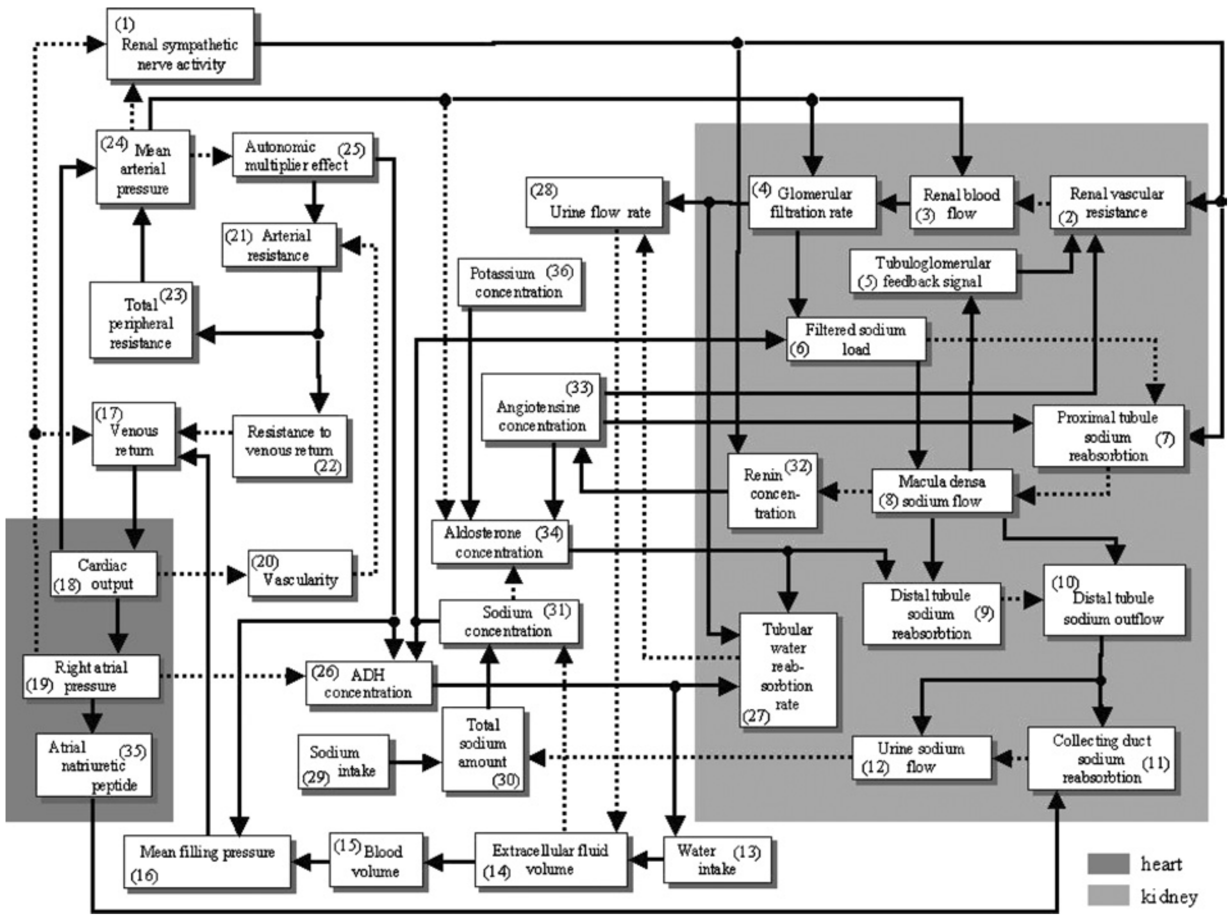


FIGURE 2. The block diagram of the basic physiological mechanisms involved in blood volume regulation. Each block expresses the dependence of a physical or physiological variable on other variables. *Continuous and dotted arrows* are used to represent the stimulating and inhibiting effects, respectively.

physiological variable on other physical or physiological variables. These blocks are briefly explained below. The mathematical relations are given in the respective equations. The normal steady-state values of the variables are given in the glossary of terms (Table 2). Blocks 1, 7, 11, 32, and 35 include novel representations or mechanisms, which are not present in the previous models

Block 1: Renal nerves from brain to kidneys (efferent renal nerves) are primarily affected by arterial pressure, cardiac pressure, and angiotensin hormone. It was shown that an increase in arterial pressure and right atrial pressure causes the renal sympathetic nerve activity to decrease in DiBona *et al.*[10,11] It was also shown that a short-term increase in the angiotensin hormone

TABLE 2. Glossary of terms.

Symbol	Name/Definition	Normal steady-state values/unit
α_{map}	Effect of mean arterial pressure on renal sympathetic nerve activity	1
α_{rap}	Effect of right atrial pressure on renal sympathetic nerve activity	1
β_{rsna}	Effect of renal sympathetic nerve activity on afferent arteriolar resistance	1
δ_{ra}	Effect of right atrial pressure on normalized antidiuretic hormone secretion rate	0
γ_{at}	Effect of angiotensin hormone concentration on fractional proximal sodium reabsorption	1
γ_{filsod}	Effect of the filtered sodium load on fractional proximal sodium reabsorption	1
γ_{rsna}	Effect of renal sympathetic nerve activity on fractional proximal sodium reabsorption	1
ε_{aum}	Autonomic multiplier effect	1
$\eta_{cd-sodreab}$	Fractional collecting duct sodium reabsorption	0.93
$\eta_{dt-sodreab}$	Fractional distal tubule sodium reabsorption	0.5
$\eta_{pt-sodreab}$	Fractional proximal sodium reabsorption	0.8
λ_{anp}	Effect of atrial natriuretic peptide on collecting duct sodium reabsorption rate	1
λ_{dt}	Effect of distal tubule sodium outflow on collecting duct sodium reabsorption rate	1
μ_{adh}	Effect of antidiuretic concentration on tubular water reabsorption rate	1
μ_{al}	Effect of aldosterone concentration on tubular water reabsorption rate	1
ν_{md-sod}	Effect of macula densa sodium flow on normalized renin secretion rate	1
ν_{rsna}	Effect of renal sympathetic nerve activity on normalized renin secretion rate	1
ξ_{at}	Effect of angiotensin hormone on normalized aldosterone secretion rate	1
$\xi_{k/sod}$	Effect of potassium to sodium concentration ratio on normalized aldosterone secretion rate	1
ξ_{map}	Effect of mean arterial pressure on normalized aldosterone secretion rate	1
Σ_{tgf}	Tubuloglomerular feedback signal	1
$\Phi_{cd-sodreab}$	Absolute collecting duct sodium reabsorption rate	1.674 mEq/min
Φ_{co}	Cardiac output	5 l/min
Φ_{dt-sod}	Distal tubule sodium outflow	1.8 mEq/min
$\Phi_{dt-sodreab}$	Absolute distal tubule sodium reabsorption rate	1.8 mEq/min
Φ_{filsod}	Filtered sodium load	18 mEq/min
Φ_{gfilt}	Glomerular filtration rate	0.125 l/min
Φ_{md-sod}	Macula densa sodium flow	3.6 mEq/min
$\Phi_{pt-sodreab}$	Absolute proximal tubular sodium reabsorption rate	14.4 mEq/min
Φ_{rb}	The renal blood flow	1.2 l/min
Φ_{sodin}	Sodium intake	0.126 mEq/min
$\Phi_{t-wreab}$	Tubular water reabsorption rate	0.124 l/min
Φ_{u}	Urine flow rate	0.001 l/min
Φ_{u-sod}	Urine sodium flow	0.126 mEq/min
Φ_{vr}	Venous return	5 l/min
Φ_{win}	Water intake	0.001 l/min
ψ_{al}	Effect of aldosterone concentration on fractional distal sodium reabsorption	1
a_{auto}	Autonomus system activity	1
a_{baro}	Baroreceptor activity	0.75
a_{chemo}	Chemoreceptor activity	0.25
C_{adh}	Antidiuretic hormone concentration	4 munits/l
C_{al}	Aldosterone concentration	85 ng/l
$\hat{C}_{anp}$	Normalized atrial natriuretic peptide concentration	1
C_{at}	Angiotensin concentration	20 ng/l
C_{gcf}	Glomerular capillary filtration coefficient	0.00781
C_{k}	Potassium concentration	5 mEq/l
$\hat{C}_{r}$	Normalized renin concentration	1
C_{sod}	Sodium concentration	144 mEq/l
K_{bar}	Coefficient relating basic arterial resistance to vascularity	16.6 mmHg min l^{-1}
K_{vd}	Vascularity destruction coefficient	0.00001

TABLE 2. Continued.

Symbol	Name/Definition	Normal steady-state values/unit
M_{sod}	Total amount of sodium	2160 mEq
N_{adh}	Normalized antidiuretic hormone concentration	1
N_{als}	Normalized aldosterone secretion rate	1
$n_{\varepsilon-dt}$	Normal value of fractional distal sodium reabsorption	0.5
$n_{\eta-cd}$	Normal value of fractional collecting duct sodium reabsorption	0.93
$n_{\eta-pt}$	Normal value of fractional proximal sodium reabsorption	0.8
N_{al}	Normalized aldosterone concentration	1
N_{rs}	Normalized renin secretion rate	1
N_{rsna}	Normalized renal sympathetic nerve activity	1
P_B	Bowman hydrostatic pressure	18 mmHg
P_f	Net filtration pressure	16 mmHg
P_{gh}	Glomerular hydrostatic pressure	62 mmHg
P_{go}	Glomerular osmotic pressure	28 mmHg
P_{ma}	Mean arterial pressure	100 mmHg
P_{mf}	Mean filling pressure	7 mmHg
P_{ra}	Right atrial pressure	0 mmHg
R_a	Arterial resistance	16.6 mmHg min l^{-1}
R_{aa}	Afferent arteriolar resistance	31.67 mmHg min l^{-1}
R_{ba}	Basic arterial resistance	16.6 mmHg min l^{-1}
R_{bv}	Basic venous resistance	3.4 mmHg min l^{-1}
R_{ea}	Efferent arteriolar resistance	51.66 mmHg min l^{-1}
rsna	Renal sympathetic nerve activity	1
R_{tp}	Total peripheral resistance	20 mmHg min l^{-1}
R_r	Renal vascular resistance	83.33 mmHg min l^{-1}
R_{vr}	Resistance to venous return	1.4 mmHg min l^{-1}
N_{adhs}	Normalized antidiuretic hormone secretion rate	1
T_{adh}	Time constant for antidiuretic hormone secretion	6 min
T_{al}	Time constant for aldosterone hormone secretion	30 min
T_r	Time constant for renin secretion	15 min
vas	Vascularity	1
vas_d	Vascularity destruction rate	0.00001
vas_f	Vascularity formation rate	0.00001
V_b	Blood volume	5 l
V_{ecf}	Extracellular fluid volume	15 l

concentration increases renal sympathetic nerve activity (rsna) in DiBona *et al.*[9] However, Lohmeier[31] has mentioned an indirect measurement showing that hypertension induced by long-term angiotensin infusion causes a decrease in rsna. Finally, in 2003 Barrett *et al.*[2] have been able to measure directly that long-term (7 days) angiotensin infusion causes a sustained increase in mean arterial pressure and a sustained decrease in rsna. Consequently, in the proposed long-term model no direct effect of angiotensin hormone on rsna has been included.

Rsna is modeled as the product of normalized rsna (N_{rsna}), the effect of mean arterial pressure on rsna (α_{map}) and the effect of right atrial pressure on rsna (α_{rap}). The relationship between mean arterial pressure and heart rate exhibits the typical adaptation behavior, such that the effect of a change in mean arterial pressure level vanishes within 1–2 days. However, in Barrett *et al.*,[2] it is reported that no similar resetting has been observed in the relationship between mean arterial pressure and rsna (i.e., baroreflex control of rsna) within 7 days after a change in the mean arterial pressure has been induced. Consequently, no adaptation mechanism has been included in the equation of α_{map}. The equations for α_{map} and α_{rap} have been obtained by curve fitting from the figures given in DiBona *et al.*[10,11] This block consists of a novel representation of rsna not given in the considered previous cardiovascular/renal models [Eqs. (2), (3), and (4)].

$$\text{rsna} = N_{rsna} \times \alpha_{map} \times \alpha_{rap} \tag{2}$$

$$\alpha_{map} = 0.5 + \frac{1.1}{1 + e^{(P_{ma} - 100\,\text{mmHg})/15\,\text{mmHg}}} \tag{3}$$

$$\alpha_{rap} = 1 - 0.008(\text{mmHg})^{-1} P_{ra} \tag{4}$$

Block 2: The renal vascular resistance (R_r) is composed of the afferent and efferent arteriolar resistances (R_{aa} and R_{ea}). The efferent arteriolar resistance (R_{ea}) has been modeled as a function of the angiotensin hormone concentration (C_{at}), the equation for which has been extracted from experimental data given by Guyton *et al.*[21] On the other hand, the afferent arteriolar resistance (R_{aa}) is modeled as the product of a constant normal value ($R_{aa-ss} = 31.67$ mmHg min l^{-1}), the effect of renal sympathetic nerve activity on afferent arteriolar resistance (β_{rsna}) and the effect of tubuloglomerular feedback signal on afferent arteriolar resistance (Σ_{tgf}). β_{rsna} is expressed as a function of rsna. The relating equations have been obtained from Guyton *et al.*[22]

$$R_r = R_{aa} + \mathrm{R}_{ea} \tag{5}$$

$$R_{aa} = R_{aa-ss} \times \beta_{rsna} \times \Sigma_{tgf} \tag{6}$$

$$\beta_{rsna} = 1.5(\mathrm{rsna} - 1) + 1 \tag{7}$$

$$R_{ea} = 0.9432 + \frac{0.1363}{0.2069 + e^{3.108 - 1.785 \log_{10} C_{at}}} \tag{8}$$

Block 3: The renal blood flow (Φ_{rb}) is obtained by dividing the mean arterial pressure (P_{ma}) by the renal vascular resistance (R_r) in accordance with Guyton's model.[22] Here it is assumed that the mean arterial pressure (P_{ma}) sufficiently well approximates the renal arterial pressure.

$$\Phi_{rb} = P_{ma}/R_r \tag{9}$$

Block 4: The relations in this block are borrowed from Guyton *et al.*[22] The glomerular filtration rate (Φ_{gfilt}) is expressed as the product of the net filtration pressure (P_f) and a constant glomerular capillary filtration coefficient (C_{gcf}). The net filtration pressure (P_f) is obtained by subtracting the sum of Bowman hydrostatic pressure (P_B) and the glomerular osmotic pressure (P_{go}) from the glomerular hydrostatic pressure (P_{gh}). The glomerular hydrostatic pressure (P_{gh}) is calculated as the difference of mean arterial pressure (P_{ma}) and the mean afferent arteriolar pressure, which is represented as the product of the renal blood flow (Φ_{rb}) and the afferent arteriolar resistance (R_{aa}). P_B and P_{go} were assumed constant at their normal steady-state values.

$$\Phi_{gfilt} = P_f \times C_{gcf} \tag{10}$$

$$P_f = P_{gh} - (P_B + P_{go}) \tag{11}$$

$$P_{gh} = P_{ma} - \Phi_{rb} \times R_{aa} \tag{12}$$

Block 5: The sodium flow sensed at macula densa gives rise to a signal, which is mainly fed back to the afferent arteriole in order to regulate the glomerular filtration rate. The tubuloglomerular feedback signal (Σ_{tgf}) was modeled as a function of macula densa sodium flow (Φ_{md-sod}).[26]

$$\Sigma_{tgf} = 0.3408 + \frac{3.449}{3.88 + e^{(\Phi_{md-sod} - 3.859)/(-0.9617)}} \tag{13}$$

Block 6: The filtered sodium load (Φ_{filsod}) represents the amount of sodium filtered from the glomerulus to the proximal tubule per unit time, and can be expressed as the product of the glomerular filtration rate (Φ_{gfilt}) and the sodium concentration (C_{sod}) as proposed by Coleman.[7]

$$\Phi_{filsod} = \Phi_{gfilt} \times C_{sod} \tag{14}$$

Block 7: Some fraction (the so-called fractional proximal sodium reabsorption, $\eta_{pt-sodreab}$) of the filtered sodium load (Φ_{filsod}) is reabsorbed from the proximal tubule to the blood. Hence the absolute proximal tubular sodium reabsorption rate ($\Phi_{pt-sodreab}$) can be represented as the product of filtered sodium load (Φ_{filsod}) and $\eta_{pt-sodreab}$ [Eq. (15)]. The fractional proximal sodium reabsorption ($\eta_{pt-sodreab}$) is affected by filtered sodium load,[7] angiotensin hormone concentration (C_{at})[7] and rsna.[8,35] Following Coleman's approach,[7] the fractional proximal sodium reabsorption ($\eta_{pt-sodreab}$) has been modeled as the product of its normal value ($n_{\eta-pt} = 0.8$) and the effects of the filtered sodium load, the angiotensin hormone concentration and rsna on fractional proximal sodium reabsorption, γ_{filsod}, γ_{at}, and γ_{rsna}, respectively, where γ_{rsna} is a novel contribution not present in Coleman's model [Eq. (16)].

An increase in filtered sodium load (Φ_{filsod}) has a decreasing influence on fractional proximal sodium reabsorption. This effect (γ_{filsod}) has been modeled by curve fitting to the piece-wise linear representation given in[7] [Eq. (17)].

An increase in angiotensin concentration (C_{at}) has an increasing influence on fractional proximal sodium reabsorption. This effect (γ_{at}) has also been modeled by curve fitting to the piece-wise linear representation given in[7] [Eq. (18)].

The results presented by Denton *et al.*[8] and Miki *et al.*[35] show that an increase in rsna has a direct increasing effect on tubular sodium reabsorption rate. On the other hand, the evidence provided by Nomora *et al.*[37] shows that rsna primarily affects the proximal tubule sodium reabsorption, hence it seems plausible to exclude the effects of rsna on sodium reabsorption at other parts of the renal tubular system. The effect of rsna on fractional proximal sodium reabsorption (γ_{rsna}) has been modeled by curve fitting to the data provided in[8] and[35] [Eq. (19)].

$$\Phi_{pt-sodreab} = \Phi_{filsod} \times \eta_{pt-sodreab} \tag{15}$$

$$\eta_{pt-sodreab} = n_{\eta-pt} \times \gamma_{filsod} \times \gamma_{at} \times \gamma_{rsna} \tag{16}$$

$$\gamma_{filsod} = 0.8 + \frac{0.3}{1 + e^{1 + [(\Phi_{filsod} - 14\,\mathrm{mEq/min})/(138\,\mathrm{mEq/min})]}} \tag{17}$$

$$\gamma_{at} = 0.95 + \frac{0.12}{1 + e^{2.6 - 1.8 \log_{10}[(C_{at})/(1\,\mathrm{ng/l})]}} \tag{18}$$

$$\gamma_{\mathrm{rsna}} = 0.5 + \frac{0.7}{1 + e^{(1-\mathrm{rsna})/2.18}} \tag{19}$$

Block 8: Since the part of the filtered sodium load (Φ_{filsod}), which is not reabsorbed into the blood, goes to macula densa, the macula densa sodium flow ($\Phi_{\mathrm{md-sod}}$) is calculated as the difference of the filtered sodium load (Φ_{filsod}) and the absolute proximal tubule sodium reabsorption rate ($\Phi_{\mathrm{pt-sodreab}}$).

$$\Phi_{\mathrm{md-sod}} = \Phi_{\mathrm{filsod}} - \Phi_{\mathrm{pt-sodreab}} \tag{20}$$

Block 9: Some fraction ($\eta_{\mathrm{dt-sodreab}}$) of the macula densa sodium flow ($\Phi_{\mathrm{md-sod}}$) is reabsorbed into the blood at the distal tubule. Thus, the absolute distal tubule sodium reabsorption rate ($\Phi_{\mathrm{dt-sodreab}}$) can be expressed as the product of the macula densa sodium flow ($\Phi_{\mathrm{md-sod}}$) and the so-called fractional distal tubule sodium reabsorption ($\eta_{\mathrm{dt-sodreab}}$). The fractional distal tubule sodium reabsorption ($\eta_{\mathrm{dt-sodreab}}$) is affected by aldosterone hormone concentration.[7] The fractional distal sodium reabsorption ($\eta_{\mathrm{dt-sodreab}}$) is represented as the product of its normal value ($n_{\varepsilon-\mathrm{dt}} = 0.5$) and the effect of aldosterone hormone concentration on fractional distal sodium reabsorption (ψ_{al}). The piece-wise linear relationship given by Uttamsingh *et al.*[40] has been modified to represent ψ_{al} as a function of the aldosterone hormone concentration ($\mathrm{C_{al}}$).

$$\Phi_{\mathrm{dt-sodreab}} = \Phi_{\mathrm{md-sod}} \times \eta_{\mathrm{dt-sodreab}} \tag{21}$$

$$\eta_{\mathrm{dt-sodreab}} = n_{\varepsilon-\mathrm{dt}} \times \psi_{\mathrm{al}} \tag{22}$$

$$\psi_{\mathrm{al}} = 0.17 + \frac{0.94}{1 + e^{[0.48-1.2\log_{10}(C_{\mathrm{al}}/(1\,\mathrm{ng/l}))]/0.88}} \tag{23}$$

Block 10: The distal tubule sodium outflow ($\Phi_{\mathrm{dt-sod}}$) is calculated as the difference of macula densa sodium flow ($\Phi_{\mathrm{md-sod}}$) and absolute distal tubule sodium reabsorption rate ($\Phi_{\mathrm{dt-sodreab}}$).

$$\Phi_{\mathrm{dt-sod}} = \Phi_{\mathrm{md-sod}} - \Phi_{\mathrm{dt-sodreab}} \tag{24}$$

Block 11: At the collecting duct some fraction ($\eta_{\mathrm{cd-sodreab}}$) of the sodium contained in the flow is reabsorbed into the blood. Hence, in accordance with,[7] the absolute collecting duct sodium reabsorption rate ($\Phi_{\mathrm{cd-sodreab}}$) is modeled as the product of the distal tubule sodium outflow ($\Phi_{\mathrm{dt-sod}}$) and so-called fractional collecting duct sodium reabsorption ($\eta_{\mathrm{cd-sodreab}}$). The fractional collecting duct sodium reabsorption ($\eta_{\mathrm{cd-sodreab}}$) can be represented as the product of its normal value ($n_{\eta-\mathrm{cd}} = 0.93$) and the effects of distal tubule sodium outflow ($\Phi_{\mathrm{dt-sod}}$) and atrial natriuretic peptide hormone concentration ($\hat{C}_{\mathrm{anp}}$) on fractional collecting duct sodium reabsorption, λ_{dt} (adapted from Coleman *et al.*[7]) and λ_{anp} (derived from Huang *et al.*[25]), respectively. The effect of atrial natriuretic peptide hormone concentration is particularly dominant at collecting duct of renal tubular system;[19] hence, this effect has not been considered at other parts of the renal tubular system. This block includes a novel representation of the effect of atrial natriuretic peptide on sodium excretion not present in previous models [Eqs. (26) and (28)].

$$\Phi_{\mathrm{cd-sodreab}} = \Phi_{\mathrm{dt-sod}} \times \eta_{\mathrm{cd-sodreab}} \tag{25}$$

$$\eta_{\mathrm{cd-sodreab}} = n_{\eta-\mathrm{cd}} \times \lambda_{\mathrm{dt}} \times \lambda_{\mathrm{anp}} \tag{26}$$

$$\lambda_{\mathrm{dt}} = 0.82 + \frac{0.39}{1 + e^{(\Phi_{\mathrm{dt-sod}}-1.6\,\mathrm{mEq/min})/2\,\mathrm{mEq/min}}} \tag{27}$$

$$\lambda_{\mathrm{anp}} = -0.1 \times \hat{C}_{\mathrm{anp}} + 1.1199 \tag{28}$$

Block 12: Sodium remaining in the collecting duct constitutes the urine sodium flow ($\Phi_{\mathrm{u-sod}}$), which can be expressed as the difference of the distal tubule sodium outflow ($\Phi_{\mathrm{dt-sod}}$) and the absolute collecting duct sodium reabsorption rate ($\Phi_{\mathrm{cd-sodreab}}$).

$$\Phi_{\mathrm{u-sod}} = \Phi_{\mathrm{dt-sod}} - \Phi_{\mathrm{cd-sodreab}} \tag{29}$$

Block 13: Although the water intake of a person can be considered as a voluntary act, hence an independent variable, there are many physiological factors that affect the level of thirst. Guyton points out the parallelism between antidiuretic hormone secretion and various stimuli known to affect thirst (e.g., high sodium concentration and hypovolemia, both of which have thirst-inducing effects, are accompanied by high antidiuretic hormone secretion).[22] In the proposed model Guyton's approach is used and water intake (Φ_{win}) is represented as a function of antidiuretic hormone concentration (C_{adh}).

$$\Phi_{\mathrm{win}} = \frac{0.008\,\mathrm{l/min}}{1 + 1.822\left(\frac{C_{\mathrm{adh}}}{1\,\mathrm{milliunits/l}}\right)^{-1.607}} - 0.0053\,\mathrm{l/min} \tag{30}$$

Block 14: The extracellular fluid volume (V_{ecf}), being the accumulation of net fluid intake rate, is calculated as the time integral of the difference between the water intake (Φ_{win}) and the urine output rate (Φ_{u}), as proposed by Guyton.[20] In the simulations the initial condition (IC) has been taken equal to the normal value of 15 l.

$$V_{\mathrm{ecf}}(t) = \mathrm{IC} + \int_0^t (\Phi_{\mathrm{win}} - \Phi_{\mathrm{u}})\,d\tau, \qquad \mathrm{IC} = 15\,\mathrm{l} \tag{31}$$

Block 15: Based on Guyton's model,[20] the blood volume (V_b) is represented as a function of extracellular fluid volume (V_{ecf}).

$$V_b = 4.5602271 + \frac{2.4312171}{1 + e^{-(V_{ecf} - 18.11278\,l)\times(0.47437\,l^{-1})}} \quad (32)$$

Block 16: Since the blood pressure is non-uniform within the circulatory system, an average variable needs to be defined. Therefore, Guyton defines mean filling pressure (P_{mf}) as "the average of all pressures in all individual segments of the systemic circulation when each of these pressures is weighted in proportion to the compliance to the compliance of the respective segment."[20] He indicates that "it can be measured by stopping the circulation and rapidly pumping blood from arteries to the veins," thus bringing the pressures in the two major compliance areas of the systemic circulation into equilibrium within a few seconds.[20] The average compliance of all systemic circulation segments is affected by the autonomous nervous system. Thus, mean filling pressure (P_{mf}) is expressed as a function of blood volume (V_b) and the autonomic multiplier effect (ε_{aum}).

$$P_{mf} = (7.436\,\text{mmHg/l} \times V_b - 30.18\,\text{mmHg}) \times \varepsilon_{aum} \quad (33)$$

Block 17: The venous return (Φ_{vr}), which is blood flow into the heart, can be calculated dividing the difference between mean filling pressure (P_{mf}) and right atrial pressure (P_{ra}) by the resistance to venous return (R_{vr}).[20]

$$\Phi_{vr} = \frac{P_{mf} - P_{ra}}{R_{vr}} \quad (34)$$

Block 18: At steady state, the input to and the output from the heart must be equal.[23] Hence, cardiac output (Φ_{co}) is taken equal to venous return (Φ_{vr}).

Block 19: The right atrial pressure (P_{ra}) is calculated as a function of the cardiac output (Φ_{co}) in accordance with the Frank-Starling Law.[20]

$$P_{ra} = 0.2787\,\text{mmHg} \times e^{\Phi_{co}\times(0.2281\ \text{min/l})} \quad (35)$$

Block 20: Vascularity gives an average measure of the number and diameter of blood vessels in the body. When there is excess blood flow in the circulatory system lasting for long periods of time, the number of tissue vessels and their diameters slowly increase to accommodate the extra flow. Similarly, when there is too little flow, "vascularity" of the tissues decreases slowly. This long-term auto-regulation of the blood vessels is accounted for by expressing the vascularity (vas) as the time integral of the net vascularity increase rate, which is the difference between vascularity formation rate (vas_f) and vascularity destruction rate (vas_d). The vascularity destruction rate (vas_d) can be taken as a constant fraction $K_{vd} = 0.00001$ (the numerical value of the related parameter k_2 in Figs. 11-5, p. 184 in Ref.[20] has been obtained via personal communication with Guyton, A.C) of the existing vascularity, while the vascularity formation rate (vas_f) can be expressed as a function of the cardiac output (Φ_{co}).[20]

$$\text{vas}(t) = \text{IC} + \int_0^t (\text{vas}_f - \text{vas}_d)\,d\tau, \quad \text{IC} = 1 \quad (36)$$

$$\text{vas}_f = \frac{11.312 \times e^{-\Phi_{co}\times(0.4799\ \text{min/L})}}{100000} \quad (37)$$

$$\text{vas}_d = \text{vas} \times K_{vd} \quad (38)$$

Block 21: The arterial resistance (R_a) is modeled as the basic arterial resistance (R_{ba}) modified by the effect of autonomous nervous system (autonomic multiplier effect, ε_{aum}), where the basic arterial resistance (R_{ba}) is inversely proportional to vascularity (vas)[20] with a constant of proportionality $K_{bar} = 16.6\,\text{mmHg min l}^{-}$ 00001 (the numerical value of the related parameter k_1 in Figs. 11-5, p. 184 in Ref.[20] has been obtained via personal communication with Guyton, A.C).

$$R_a = R_{ba} \times \varepsilon_{aum} \quad (39)$$

$$R_{ba} = K_{bar}/\text{vas} \quad (40)$$

Block 22: The resistance to venous return is defined as "the average resistance from each segment of the systemic circulation to the right atrium when each of these resistances is weighted in proportion to the compliance of the respective segment."[16] The empirical formula borrowed from Guyton's model[20] expresses the resistance to venous return (R_{vr}) as a function of a constant basic venous resistance (R_{bv}) and the arterial resistance (R_a).

$$R_{vr} = (8\,R_{bv} + R_a)/31 \quad (41)$$

Block 23: The total peripheral resistance (R_{tp}) is calculated as the sum of arterial resistance (R_a) and constant basic venous resistance (R_{bv}).[20]

$$R_{tp} = R_a + R_{bv} \quad (42)$$

Block 24: Since the fluctuation between systolic and diastolic pressures is ignorable from the perspective of long-term cardiovascular dynamics, it is reasonable to use their mean, i.e., mean arterial pressure. The mean arterial pressure (P_{ma}) is calculated as the product of cardiac output (Φ_{co}) and the total peripheral resistance (R_{tp}). Angiotensin has a strong direct vasoconstrictor effect, which persists only for 1 or 2 min, while its indirect influence of increasing sodium and water reabsorption from the kidneys becomes much more dominant in the long term.[19] Consequently, in this long-term model only the indirect influence of angiotensin has been included.

$$P_{\mathrm{ma}} = \Phi_{\mathrm{co}} \times R_{\mathrm{tp}} \tag{43}$$

Block 25: Guyton *et al.*[20] have represented the effect of the autonomous nervous system on various variables in terms of a normalized variable, the autonomic multiplier effect ($\varepsilon_{\mathrm{aum}}$). The autonomic multiplier effect ($\varepsilon_{\mathrm{aum}}$) is composed of chemoreceptor (a_{chemo}) and baroreceptor (a_{baro}) activities, which are functions of the mean arterial pressure (P_{ma}) (expressed via an intermediate variable a_{auto}). As opposed to the chemoreceptor activity, the baroreceptor activity (a_{baro}) exhibits an adaptation behavior accounted for by the integral equation, where the factor 0.0000667 min^{-1} is the inverse of the adaptation time constant.

$$\varepsilon_{\mathrm{aum}} = a_{\mathrm{chemo}} + a_{\mathrm{baro}} \tag{44}$$

$$a_{\mathrm{auto}} = 3.079\, e^{-P_{\mathrm{ma}} \times 0.011\,\mathrm{mmHg}^{-1}} \tag{45}$$

$$a_{\mathrm{chemo}} = 1/4 a_{\mathrm{auto}} \tag{46}$$

$$a_{\mathrm{baro}}(t) = 3/4 \left\{ a_{\mathrm{auto}} - (0.0000667\ \overset{-1}{\mathrm{min}}) \times \left[\int_0^t (a_{\mathrm{baro}} - 1)\, d\tau \right] \right\} \tag{47}$$

Block 26: The antidiuretic hormone concentration (C_{adh}) can be expressed as the product of the normalized antidiuretic hormone concentration (N_{adh}) and normal value of antidiuretic hormone concentration (4 milli-units/l) [Eq. (48)].

Equation (49) accounts for the regulation mechanism, by which the normalized antidiuretic hormone concentration (N_{adh}) is regulated to a normalized antidiuretic hormone secretion rate (N_{adhs}). Here the time constant T_{adh} is taken as 6 min[16] and the initial condition (*IC*) as equal to the normal value = 1.

As indicated in[18] and used in,[22] the antidiuretic hormone secretion rate is mainly determined by the sodium concentration (C_{sod}), the autonomic multiplier effect ($\varepsilon_{\mathrm{aum}}$) and the effect of right atrial pressure (δ_{ra}) in an additive manner Eq. (50)]. Since C_{sod} contributes to the antidiuretic hormone secretion rate from a certain concentration level (taken as 141 mEq/l) onwards (when sodium concentration is slightly below and above than its normal value, antidiuretic hormone is secreted), in this block C_{sod} has been limited to values above this level. This lower bound corresponds to a value 3 mEq/l below the normal sodium concentration of 144 mEq/l, where 3 mEq/l is a value borrowed from.[22] Similarly, since blood volume decrease below its normal value, hence arterial pressure decrease and consequently increase of $\varepsilon_{\mathrm{aum}}$ above its normal value 1 contributes to the antidiuretic hormone secretion rate, in this block $\varepsilon_{\mathrm{aum}}$ has been limited to values above 1.

The effect of right atrial pressure (δ_{ra}) is not a simple function of the right atrial pressure (P_{ra}) but also involves a self adaptation mechanism accounted for in Eq. (51) and borrowed from.[22]

$$C_{\mathrm{adh}} = 4\,\mathrm{milli-units/l} \times N_{\mathrm{adh}} \tag{48}$$

$$N_{\mathrm{adh}}(t) = \frac{1}{T_{\mathrm{adh}}} \int_0^t (N_{\mathrm{adhs}} - N_{\mathrm{adh}})\, d\tau + \mathrm{IC}, \quad \mathrm{IC} = 1 \tag{49}$$

$$N_{\mathrm{adhs}} = \left[\frac{(C_{\mathrm{sod}} - 141\,\mathrm{mEq/l})}{1\,\mathrm{mEq/l}} + (\varepsilon_{aum} - 1) - \delta_{\mathrm{ra}} \right] / 3 \tag{50}$$

$$\delta_{\mathrm{ra}}(t) = (0.2\,\mathrm{mmHg}^{-1}) P_{ra} - (0.0007\ \overset{-1}{\mathrm{min}}) \left[\int_0^t \delta_{\mathrm{ra}}\, d\tau \right] \tag{51}$$

Block 27: In this block the tubular water reabsorption rate ($\Phi_{\mathrm{t-wreab}}$) is calculated in terms of the effects of aldosterone concentration and antidiuretic hormone concentration (μ_{al} and μ_{adh}) and the glomerular filtration rate (Φ_{gfilt}) as proposed by Guyton.[22] However, the relations for μ_{al} and μ_{adh} have been derived from Uttamsingh's model.[40]

$$\Phi_{\mathrm{t-wreab}} = 0.025\,\mathrm{l/min} - \frac{0.001\,\mathrm{l/min}}{\mu_{\mathrm{al}} \times \mu_{\mathrm{adh}}} + 0.8 \Phi_{\mathrm{gfilt}} \tag{52}$$

$$\mu_{\mathrm{al}} = 0.17 + \frac{0.94}{1 + e^{[0.48 - 1.2\, \log_{10}(C_{\mathrm{al}}/(1\,\mathrm{ng/l}))]/0.88}} \tag{53}$$

$$\mu_{\mathrm{adh}} = 0.37 + \frac{0.8}{1 + e^{0.6 - 3.7\, \log_{10}(C_{\mathrm{adh}}/(1\,\mathrm{mEq/l}))}} \tag{54}$$

Block 28: The urine flow rate (Φ_{u}) is calculated as the difference between the glomerular filtration rate (Φ_{gfilt}) and the tubular water reabsorption rate ($\Phi_{\mathrm{t-wreab}}$). In the simulations, Φ_{u} has been confined to values above 0.0003 l/min as used in Guyton's model.[22] Such a lower bound for the urine flow rate has been included in order to guarantee the excretion of the daily "minimum obligatory urine volume" (~0.5 l/day) needed for getting rid of the waste products of the metabolism as indicated in.[19]

$$\Phi_{\mathrm{u}} = \Phi_{\mathrm{gfilt}} - \Phi_{\mathrm{t-wreab}} \tag{55}$$

Block 29: The sodium intake (Φ_{sodin}) has been taken as an independent variable with a normal steady-state value of 0.126 mEq/min.

Block 30: The total amount of sodium (M_{sod}) is represented as the time integral of the net sodium intake rate, i.e., the difference between sodium intake (Φ_{sodin}) and urine sodium flow ($\Phi_{\text{u-sod}}$). The initial condition (*IC*) is taken equal to the normal value = 2160 mEq.

$$M_{\text{sod}}(t) = \text{IC} + \int_0^t (\Phi_{\text{sodin}} - \Phi_{\text{u-sod}})d\tau, \quad \text{IC} = 2160\,\text{mEq} \tag{56}$$

Block 31: Sodium concentration (C_{sod}) is expressed as the ratio of the total amount of sodium (M_{sod}) to the extracellular fluid volume (V_{ecf}).

$$C_{\text{sod}} = M_{\text{sod}}/V_{\text{ecf}} \tag{57}$$

Block 32: This block represents the mechanism, by which the normalized renin concentration ($\hat{C}_{\text{r}}$) is regulated to a normalized renin secretion rate (N_{rs}), in a similar way Guyton has represented the regulation mechanisms of other hormones. The factors that determine this normalized renin secretion rate (N_{rs}) include macula densa sodium flow ($\Phi_{\text{md-sod}}$), renal sympathetic nerve activity (rsna) and the blood pressure sensed by the baroreceptors at the juxtaglomerular complex.[12] However, Guyton has observed that the baroreceptor mechanism has a negligible direct effect on renin secretion.[21] Hence, in the proposed model the normalized renin secretion rate (N_{rs}) is calculated in terms of the effect of macula densa sodium flow on normalized renin secretion rate ($\nu_{\text{md-sod}}$) and effect of renal sympathetic nerve activity on normalized renin secretion rate (ν_{rsna}). T_{r} represents the time constant for renin secretion and is taken as 15 min.[21] The initial condition (*IC*) is taken equal to the normal value, i.e., 1. This block includes a novel representation of the rsna effect on renin secretion not present in previous models [Eq. (61)].

$$\hat{C}_{\text{r}}(t) = \frac{1}{T_{\text{r}}}\int_0^t (N_{\text{rs}} - \hat{C}_{\text{r}})d\tau + \text{IC}, \qquad \text{IC} = 1 \tag{58}$$

$$N_{\text{rs}} = \nu_{\text{md-sod}} \times \nu_{\text{rsna}} \tag{59}$$

$$\nu_{\text{md-sod}} = 0.2262 + \frac{28.04}{11.56 + e^{(\Phi_{\text{md-sod}} - 1.667\,\text{mEq/min})/(0.6056\,\text{mEq/min})}} \tag{60}$$

$$\nu_{\text{rsna}} = 1.89 - \frac{2.056}{1.358 + e^{(\text{rsna} - 0.8667)}} \tag{61}$$

Block 33: The angiotensin concentration (C_{at}) is assumed to be proportional to the normalized renin concentration ($\hat{C}_{\text{r}}$).[7]

$$C_{\text{at}} = 20\,\text{ng/l} \times \hat{C}_r \tag{62}$$

Block 34: This block represents the mechanism, by which the normalized aldosterone concentration (N_{al}) is regulated to the normalized aldosterone secretion rate (N_{als}). The time constant T_{al} governing the regulation of aldosterone concentration is taken as 60 min, as used by Guyton *et al.*[22] The normalized aldosterone secretion rate (N_{als}) is calculated in terms of the effects of potassium to sodium concentration ratio, mean arterial pressure and angiotensin concentration ($\xi_{\text{k/sod}}$, ξ_{map}, and ξ_{at}, respectively) as suggested by Guyton *et al.*[22] Here, in spite of its important contribution to aldosterone secretion, the potassium concentration C_{K} has been assumed to remain constant at its normal level of 5 mEq/l, since the objective of this model is to investigate the relationship between renal sympathetic activity, arterial pressure and sodium excretion. Although no direct effect of mean arterial pressure on aldosterone secretion rate has been reported in general, Guyton indicates that a direct effect may exist in cases of hypotension.[21] Based on this, the effect of the mean arterial pressure on aldosterone secretion (ξ_{map}) has been assumed as 1 (no effect) for mean arterial pressures above normal, while for hypotension cases ($P_{\text{ma}} < 100$ mmHg) the relation provided by Guyton *et al.*[22] has been adopted. The effect of angiotensin hormone concentration on aldosterone secretion rate (ξ_{at}) has been obtained by smoothing the piece-wise linear representation provided by Uttamsingh *et al.*[40]

In the simulations aldosterone concentration (C_{al}) has been limited to a range between 1–500 ng/l in order to guarantee the stability of the overall algorithm. Similarly, $\xi_{\text{k/sod}}$ has been confined to non-negative values only.

Aldosterone concentration (C_{al}) is equal to the product of the normalized aldosterone concentration (N_{al}) and the normal value of aldosterone concentration (85 ng/l). The initial condition (IC) is taken equal to the normal value, i.e., 1.

$$C_{\text{al}} = N_{\text{al}} \times 85\,\text{ng/l} \tag{63}$$

$$N_{\text{al}}(t) = \frac{1}{T_{\text{al}}}\int_0^t (N_{\text{als}} - N_{\text{al}})d\tau + \text{IC}, \qquad \text{IC} = 1 \tag{64}$$

$$N_{\text{als}} = \xi_{\text{k/sod}} \times \xi_{\text{map}} \times \xi_{\text{at}} \tag{65}$$

$$\xi_{\text{k/sod}} = \frac{C_{\text{K}}/C_{\text{sod}}}{0.003525} - 9 \tag{66}$$

$$\xi_{\text{map}} = \begin{cases} 69.03\,e^{-(0.0425\,\text{mmHg}^{-1})\times P_{\text{ma}}} & \text{if } P_{\text{ma}} \le 100\,\text{mmHg} \\ 1 & \text{if } P_{\text{ma}} > 100\,\text{mmHg} \end{cases} \tag{67}$$

$$\xi_{\text{at}} = 0.4 + \frac{2.4}{1 + e^{[2.82 - 1.5 \log_{10}(C_{\text{at}}/(1\,\text{ng/l}))]/0.8}} \quad (68)$$

Block 35: The normalized atrial natriuretic peptide concentration ($\hat{C}_{\text{anp}}$) is calculated as a function of the right atrial pressure (P_{ra}) in accordance with Lohmeier *et al.*[33] For human beings the actual value at normal conditions is about 36 ng/l.

$$\hat{C}_{\text{anp}} = 7.427 - \frac{6.554}{1 + e^{(P_{\text{ra}} - 3.762\,\text{mmHg})/(1\,\text{mmHg})}} \quad (69)$$

Block 36: This block represents the potassium concentration, which in this model has been assumed to remain constant at its normal value ($C_{\text{k}} = 5$ mEq/l).

The curve-fitting tool of Matlab has been used when obtaining non-linear equations from the findings in the literature represented in terms of piecewise linear equations or in graphical form.

The proposed cardiovascular system model has been implemented using Matlab/Simulink. Runge Kutta 4 numerical method with fixed step size of 10 min has been used to solve the nonlinear differential equations in the model.

Results of the sensitivity analysis performed by changing nine parameters one at a time by $\pm 10\%$ of their nominal values are summarized in Table 3. The resulting changes in the steady-state values of the 15 physiological variables are in general relatively small compared to the $\pm 10\%$ parameter change. The largest parameter sensitivity (10.9%) is observed in the renal blood flood as a response to a -10% change in the steady-state efferent arteriolar resistance, while all other changes remain below 10% in magnitude. Thus it can be said that the proposed model is quite robust with respect to parameter changes.

RESULTS AND DISCUSSION

Due to various reasons the evaluation of the validity of the proposed overall model is a non-trivial issue. The model consists of blocks, which are based on open-loop experimental data and are put together in order to function as a closed-loop system. As far as comparisons to real observations are concerned, we are confined to the limited amount of closed-loop clinical and experimental data available in the literature. Due to the difficulty of long-term experiments, observations and measurements, also the amount of long-term data is rather small.

Some clinical and experimental data in the literature are not given in terms of the same units. In order to provide a common basis for fair comparison such data have been first expressed as percentages of the normal steady-state values as given in the related publication, and then converted into the units used in the simulations taking the normal steady-state values of the model as a basis. Since in practice physiological variables are considered as normal within certain ranges, the mean values of such ranges have been taken as normal steady-state values. Experimental results (represented by dots in Figs. 3 and 4) have been converted into the units used in the simulations

Behavior Under Varying Sodium Intake in Healthy Subjects

Varying sodium intake is a good method for testing whether of the closed-loop model of the blood volume and pressure regulation mechanisms work properly, because a change in the sodium intake can be considered as an indirect way of causing long-term blood volume changes. Furthermore among all variables considered in the model, sodium intake is one of the few, which can be approximated as an independent variable.

The publication of Feng *et al.*[16] has been chosen as a basis of comparison, because this publication reports the results of a long-term experiment involving both increase and decrease in sodium intake. The experiment consists of high sodium intake (approximately by 200%) during the first 5 days and low sodium intake (approximately 94%) for the next 5 days in healthy human subjects [Fig. 3(A)]. As expected, extracellular fluid volume [Fig. 3(B)] and blood volume [Fig. 3(C)] increase during increasing of salt intake. An increase in the blood volume stimulates blood pressure and cardiac receptors, therefore reflex rsna inhibition is expected[2] [Fig. 3(D)]. As expected, an increase in the blood volume increases the right atrial pressure, and hence also the expansion of the right atrium causes secretion of atrial natriuretic peptide [Fig. 3(E)]. An increase in sodium intake causes the filtered sodium load and the macula densa sodium flow to increase. The increase in macula densa sodium flow and rsna decrease cause to decrease the renin secretion rate, therefore renin concentration decrease [Fig. 3(F)]. The decrease of renin concentration causes a decrease in angiotensin and aldosterone concentrations [Fig. 3(G)]. As a result of the decrease of angiotensin and aldosterone concentrations, the renal tubular sodium reabsorption decreases. In addition to this, increasing of atrial natriuretic peptide decreases the tubular sodium reabsorption. Consequently, the sodium excretion increases [Fig. 3(H)] during sodium loading [Fig. 3(A)]. Hence, in the long run the difference between sodium intake and sodium urine goes to zero [Fig. 3(I)] and the sodium balance is established. The model also predicts that the difference between water intake and urine volume will go to zero [Fig. 3(J)] establishing the water balance. The increase in sodium intake increases the antidiuretic hormone concentration, as expected. During the subsequent 5 days of sodium intake reduction compensatory mechanisms work in the opposite direction.

All together, the cooperation of several control mechanisms results in a relatively small change in the sodium concentration [Fig. 3(L)], in spite of the large variation in sodium intake.

TABLE 3. Sensitivity analysis of proposed model showing the percentage change of the steady-state value of each variable due to a +10% (upper) and −10% (lower) change of a parameter.

Output variables, the steady-state percentage change of the steady-state values of which are observed	Parameters changed by ±10% of their nominal values (one at a time)								
	N_{rsna}	R_{ea}	R_{aa-ss}	P_B	P_{go}	C_{gcf}	K_{vd}	K_{bar}	R_{bv}
Mean arterial pressure	2.6 −2.9	−1.6 1.8	1.6 −1.8	1.3 −1.3	2.1 −2.1	−1.1 1.3	0.05 −0.05	0.2 −0.3	0.05 −0.06
Glomerular filtration rate	0.4 −0.43	0.3 −0.3	−0.3 0.3	−0.2 0.2	−0.4 0.4	0.2 −0.2	0.02 −0.02	0.09 −0.1	0.02 −0.02
Renin concentration	3.6 −5.3	−0.9 0.2	0.2 −1	0.2 −0.8	0.1 −1.2	−0.6 0.2	−0.6 0.6	−2.5 3.6	−0.6 0.6
Extracellular fluid volume	0.88 −0.98	−0.5 0.6	0.5 −0.6	0.4 −0.4	0.6 −0.7	−0.3 0.4	−0.1 0.1	−0.8 0.9	−0.3 0.3
Urinary sodium flow	0.1 −0.12	−0.06 0.07	0.06 −0.07	0.05 −0.05	0.08 −0.08	−0.04 0.05	0.1 −0.1	−0.2 0.2	−0.05 0.05
Sodium concentration	0.08 −0.08	0.06 −0.06	−0.06 0.07	−0.04 0.05	−0.07 0.08	0.04 −0.04	0.003 −0.003	0.02 −0.02	0.004 −0.005
Total sodium amount	0.96 −1	−0.4 0.5	0.4 −0.5	0.3 −0.4	0.6 −0.6	−0.3 0.3	−0.1 0.1	−0.8 0.9	−0.3 0.3
Arterial resistance	0.2 −0.27	−0.1 0.1	0.1 −0.1	0.1 −0.1	0.1 −0.2	−0.1 0.1	1.1 −1.1	7.5 −7.8	−0.4 0.4
Cardiac output	2.5 −2.6	−1.5 1.7	1.5 −1.7	1.2 −1.2	1.9 −1.9	−1 1.2	−0.9 0.9	−5.6 6.6	−1.2 1.2
Renal blood flow	−0.03 0.1	−8.9 10.9	−0.1 0.1	2.8 −2.7	4.3 −4.3	−2.2 2.7	0.03 −0.03	0.1 −0.1	0.02 −0.03
Afferent arteriolar resistance	7 −7.7	4.7 −5.2	4.8 −5.2	−3.8 3.9	−5.8 6.2	3.2 −3.7	0.1 −0.1	0.5 −0.6	0.1 −0.1
Renal vascular resistance	2.7 −3	8 −8.2	1.8 −1.9	−1.4 1.4	−2.1 2.2	1.1 −1.3	0.02 −0.02	0.1 −0.1	0.02 −0.03
Absolute proximal tubular sodium reabsorption rate	0.77 −0.9	0.3 −0.4	−0.4 0.4	−0.3 0.3	−0.5 0.4	0.2 −0.3	−0.01 0.02	−0.06 0.08	−0.01 0.01
Absolute distal tubule sodium reabsorption rate	−0.7 1.7	0.9 −0.5	−0.4 1	−0.4 0.7	−0.5 1.2	0.6 −0.4	0.1 −0.1	0.6 −0.9	0.1 −0.1
Absolute collecting duct sodium reabsorption rate	−0.49 0.5	0.2 −0.3	−0.3 0.3	−0.2 0.2	−0.3 0.3	0.1 −0.2	0.2 −0.2	1 −1.3	0.2 −0.2

Note. N_{rsna}: Normalized renal sympathetic nerve activity; R_{ea}: Steady-state value of efferent arteriolar resistance; R_{aa-ss}: Steady-state value of afferent arteriolar resistance; P_B: Bowman hydrostatic pressure; P_{go}: Glomerular osmotic pressure; C_{gcf}: Glomerular capillary filtration coefficient; K_{vd}: Vascularity destruction coefficient; K_{bar}: Coefficient relating basic arterial resistance to vascularity; R_{bv}: Basic venous resistance.

Another publication, which has been used to check the closed-loop performance of the proposed model in healthy humans, is by Barba *et al.*[1] The experiment presented in this publication consists of a 3 days decrease of sodium intake (approximately by 62.5%) [Fig. 4(A)]. The variables reported in this publication include mean arterial pressure [Fig. 4(C)], glomerular filtration rate [Fig. 4(E)], absolute proximal sodium reabsorption [Fig. 4(N)], fractional proximal sodium reabsorption [Fig. 4(M)], absolute distal sodium reabsorption rate [Fig. 4(P)], fractional distal sodium reabsorption [Fig. 4(O)] and sodium concentration [Fig. 4(R)]. The simulation results are superposed on the respective Figs. as solid lines. Furthermore; rsna [Fig. 4(B)], macula densa sodium flow [Fig. 4(F)], tubulo-glomerular feedback signal [Fig. 4(N)], afferent arteriolar resistance [Fig. 4(J)], renin concentration [Fig. 4(H)], angiotensin

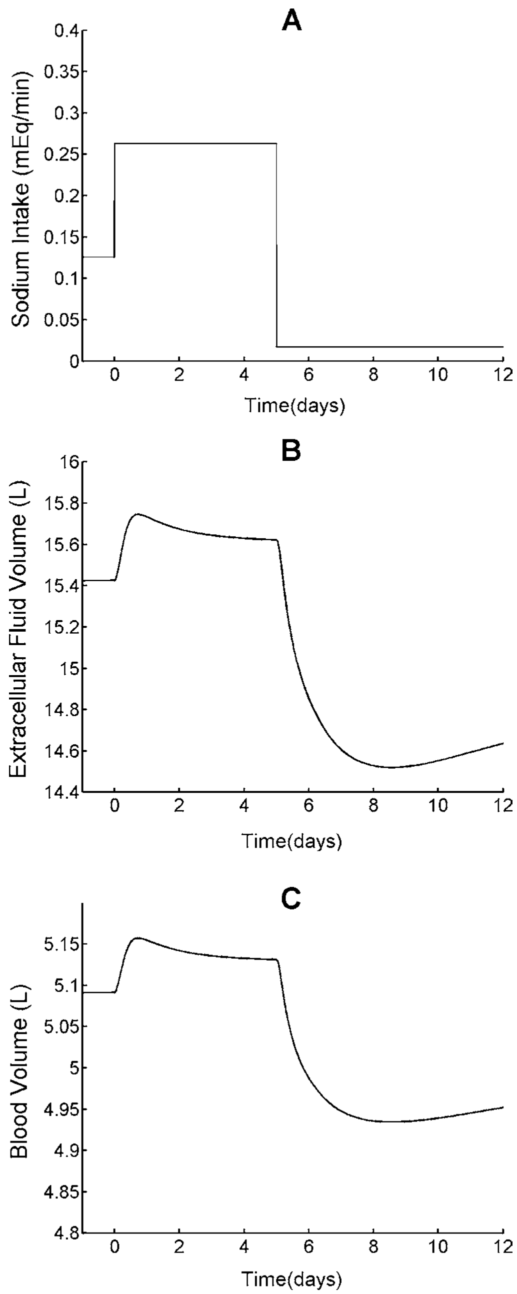


FIGURE 3. Results of the simulation of the experiment presented in Feng *et al.*[14] with 5 days of sodium loading of 200% and 5 days of sodium depletion of 94%. Simulation results are represented by *solid lines*, while experimental data scaled to match the steady-state values of the respective variables as used in the simulations are shown by *dots*.

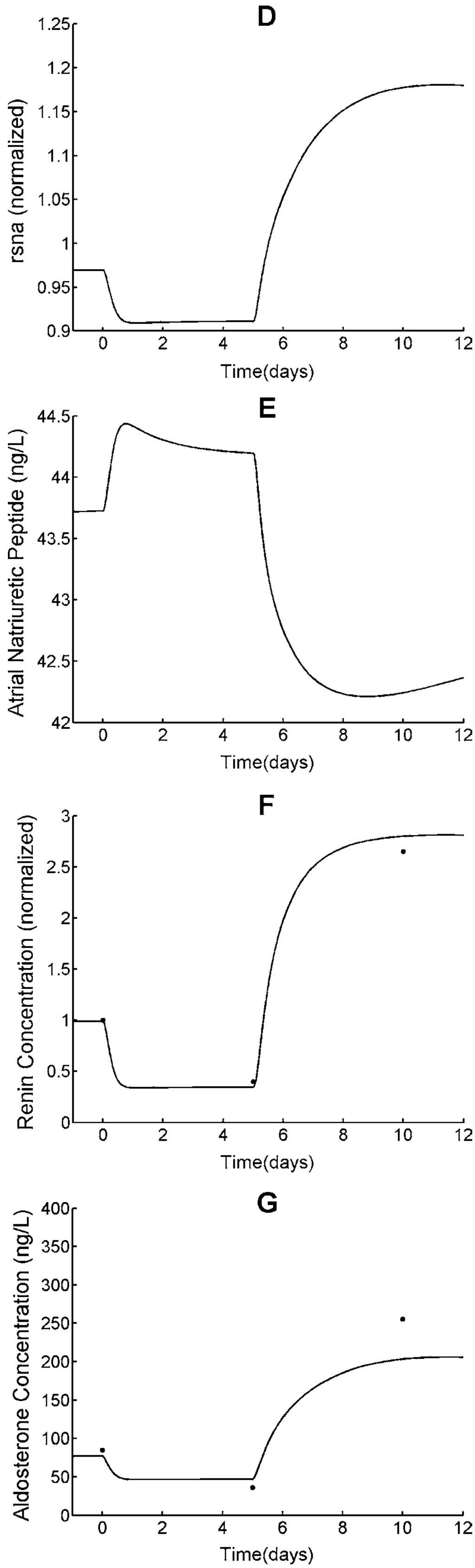


FIGURE 3. Continued.

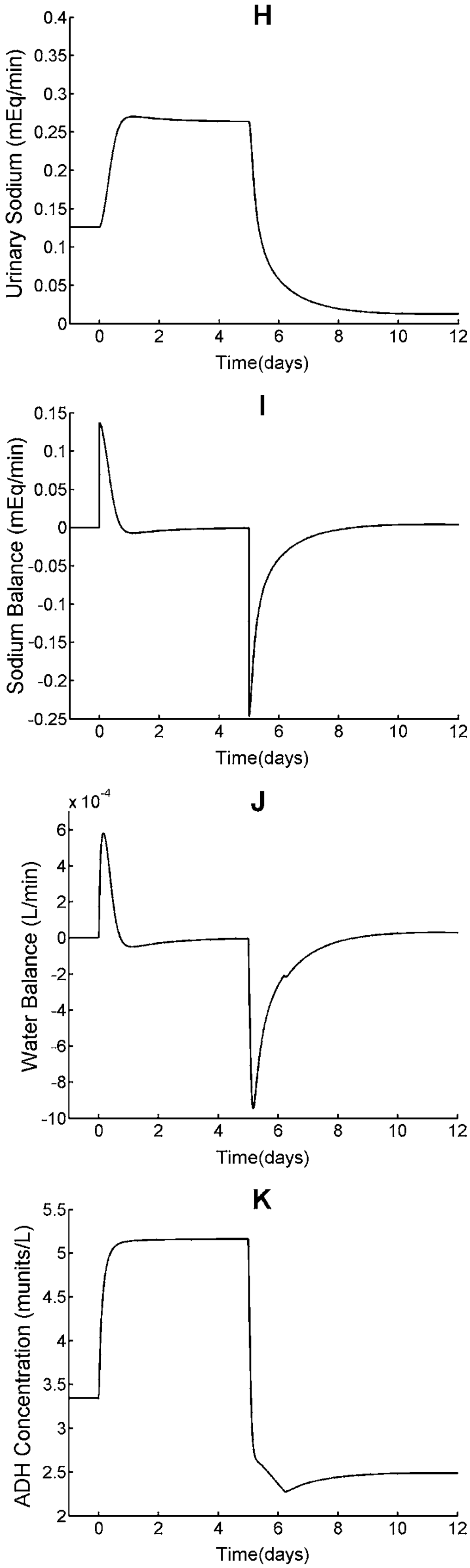


FIGURE 3. Continued.

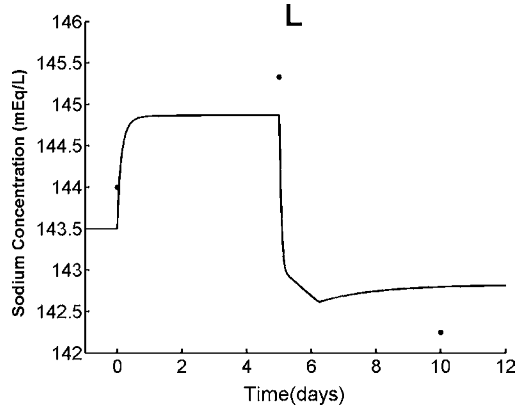


FIGURE 3. Continued.

concentration [Fig. 4(I)], efferent arteriolar resistance [Fig. 4(K)], glomerular hydrostatic pressure [Fig. 4(P)] and renal blood flow [Fig. 4(L)] as obtained from the simulations are also given, although Barba *et al.*[1] do not report the outcomes of these variables. These results demonstrate that the model exhibits the following expected behaviors:

(i) Decrease in sodium intake reduces the extracellular fluid volume and consequently also the mean arterial pressure.
(ii) Decrease in mean arterial pressure increases rsna.
(iii) Increase in rsna contributes to the increase of renin secretion and tubular sodium reabsorption.
(iv) Decrease in mean arterial pressure also reduces the glomerular filtration rate.
(v) Decrease in the glomerular filtration rate reduces the macula densa sodium flow.
(vi) Decrease in the macula densa sodium flow reduces the tubulo-glomerular feedback signal.
(vii) Decrease in the tubulo-glomerular feedback signal reduces the afferent arteriolar resistance (vasodilatation of the afferent arteriole).
(viii) The regulating effect of the decreasing afferent arteriolar resistance causes a smaller decrease in the glomerular hydrostatic pressure than would be otherwise.
(ix) Consequently, also the glomerular filtration rate decreases less than it would do without the regulating effect of the decreasing afferent arteriolar resistance.
(x) Decrease in the macula densa sodium flow causes an increase in the renin concentration and consequently an increase in the angiotensin concentration.
(xi) Increase in the angiotensin concentration causes an increase in the efferent arteriolar resistance (vasoconstruction).
(xii) The regulating effect of the increasing efferent arteriolar resistance causes a smaller decrease in the glomerular hydrostatic pressure than would be otherwise.

(xiii) Increase in the angiotensin concentration causes an increase in the fractional proximal sodium reabsorption, and consequently also an increase in the absolute proximal sodium reabsorption.

(xiv) Increase in the angiotensin concentration also causes an increase in the fractional distal sodium reabsorption by increasing the aldosterone concentration.

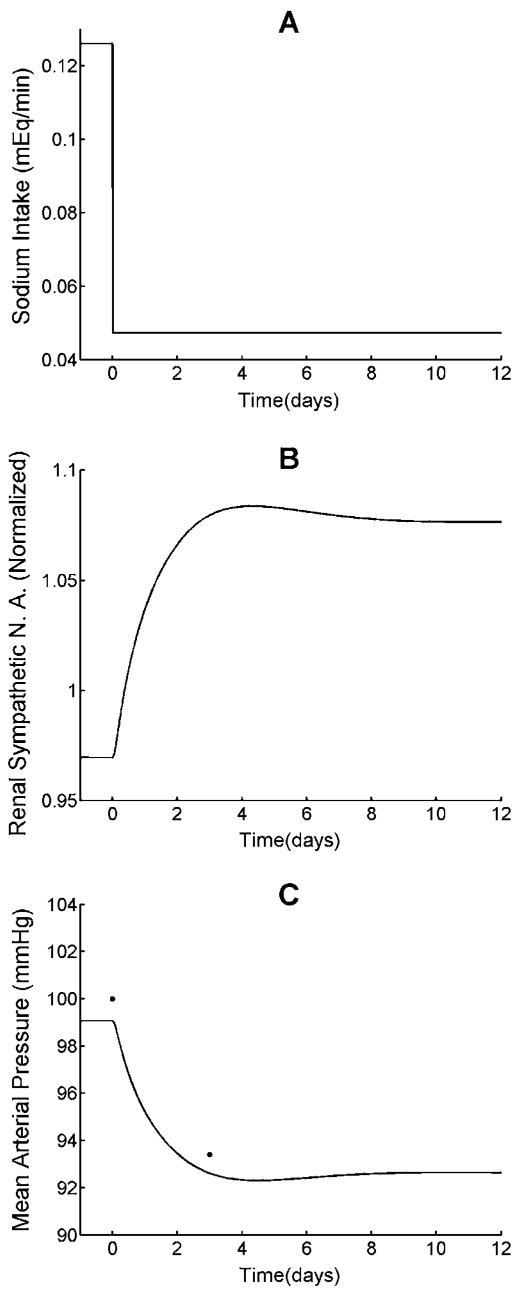


FIGURE 4. Results of the simulation of the experiment presented in Barba *et al.*[1] with 3 days of sodium intake reduction of 62.5%. Simulation results are represented by *solid lines*, while experimental data scaled to match the steady-state values of the respective variables as used in the simulations are shown by *dots*.

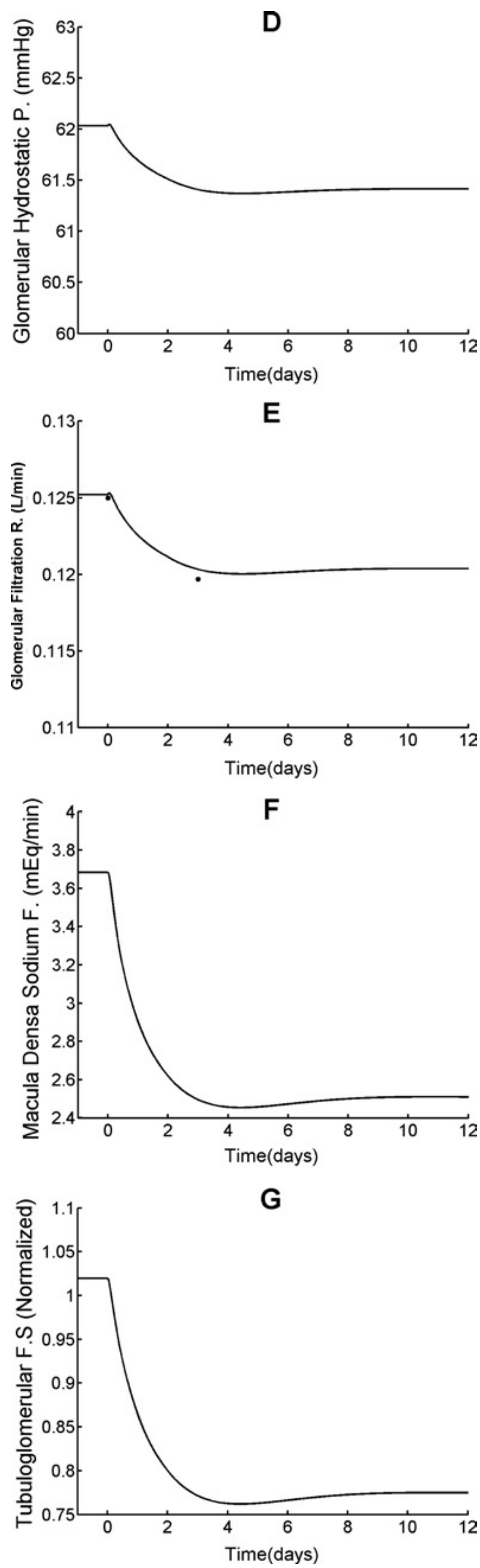
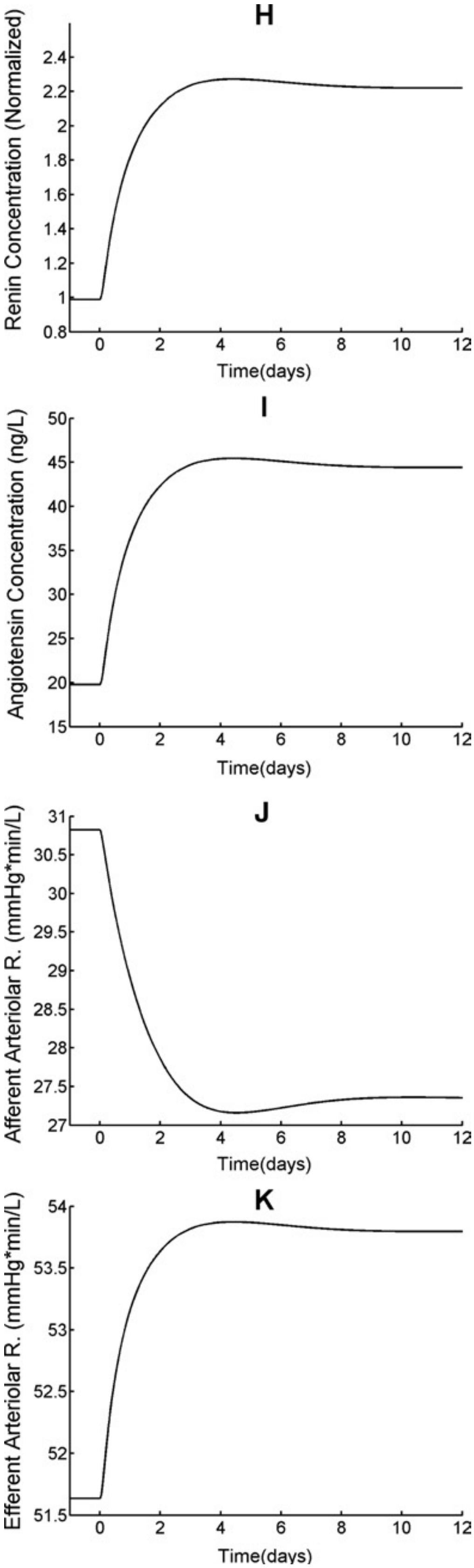


FIGURE 4. Continued.



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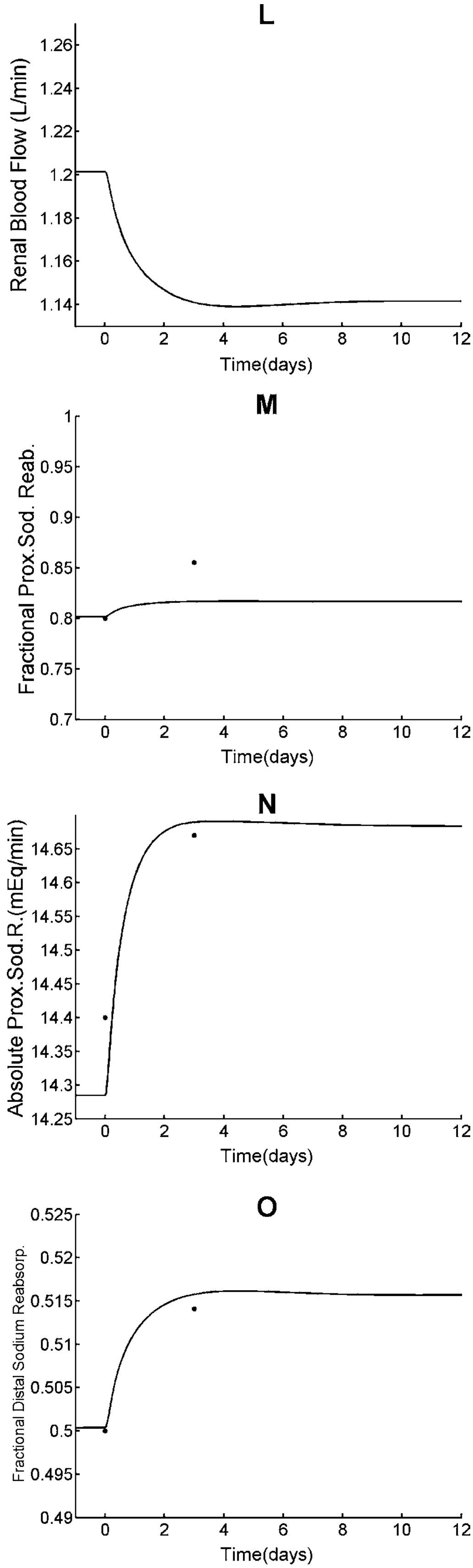

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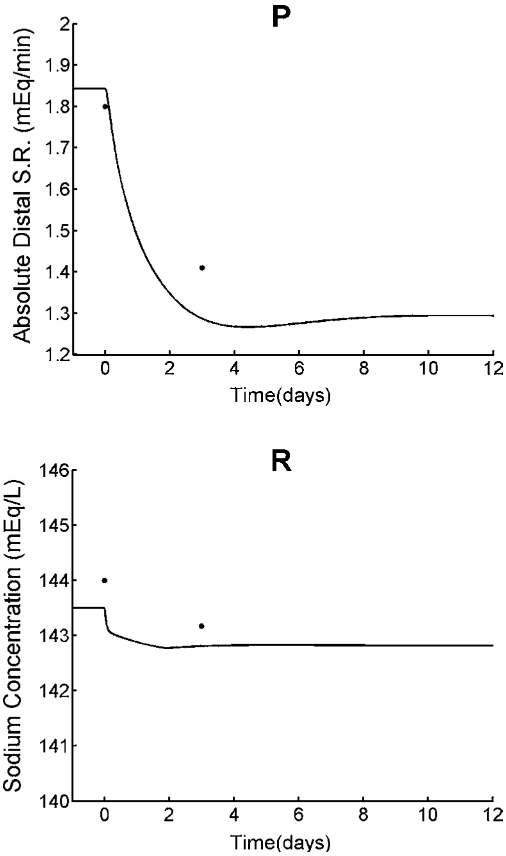

FIGURE 4. Continued.

(xv) Increase in the fractional distal sodium reabsorption reduces the absolute distal sodium reabsorption.

(xvi) All these feedback effects regulate the sodium concentration such that it drops only slightly as a response to a large decrease in the sodium intake.

Behavior Under Increased RSNA

Hypertension, congestive heart failure, nephrotic syndrome and cirrhosis are known to involve an increased basal rsna.[12] Various sources in the literature report that a short-term increase in rsna gives rise to an increase in renin secretion and tubular sodium reabsorption.[8,12,35] But there are very few publications on the effects of long-term increase in rsna on kidney functions.

One of them is a direct animal experiment, which has shown that a long-term (chronical) increase in rsna causes sustained elevation in arterial pressure.[29]

Another publication[38] reports about an attempt to approximate chronic rsna stimulation by norepinephrine infusion (norepinephrine is used as an indirect index of rsna) into the renal artery of dogs. This experiment has resulted in a short-term decrease in glomerular filtration rate, renal blood flow, sodium excretion followed by a return to control values; however, arterial pressure and plasma renin activity have remained at an increased level during 7 days.

Additionally, in Braith *et al.*[4] increases in blood volume and in the concentrations of renin, angiotensin, aldosterone, and atrial natriuretic peptide are reported for heart transplantation cases (cardiac denervation). This behavior is explained as follows in Braith *et al.*[5] "*Cardiorenal reflex mechanism of volume regulation is abnormal in heart transplant recipients. Ablation of cardiac afferent input should disinhibit the cardiorenal reflex, increase renal nerve activity and renal vascular resistance, reduce glomerular filtration rate and increase vasopressin, plasma renin activity, angiotensin II, and aldosterone. These effects should impair the ability to excrete salt and thereby increase blood pressure.*"

Since the above mentioned publications report the increase in rsna only in a qualitative manner, they constitute only a qualitative basis for checking how well the proposed model accounts for the effects of rsna increase [Fig. 5(A)] during normal sodium intake. Simulation results obtained by increasing rsna to a higher level exhibit convergence to an elevated level in mean arterial pressure [Fig. 5(B)] (in qualitative accordance with Kottke *et al.*[29] and Reinhart *et al.*[38]), blood volume [Fig. 5(F)] (in qualitative accordance with Braith *et al.*[4]), atrial natriuretic peptide [Fig. 5(G)] (in qualitative accordance with Braith *et al.*[4]). Although not measured in the aforementioned works, an increase in total sodium [Fig. 5(E)] and afferent arteriolar resistance [Fig. 5(H)] is theoretically expected due to the increase in rsna. Similarly, an initial decrease in glomerular hydrostatic pressure due to increase in afferent arteriolar resistance is expected. However, this decrease is followed by an increase due to the increase in blood volume. Since the increase in the blood volume results in an increase in the mean arterial pressure, the glomerular hydrostatic pressure increases above the control level and thus the glomerular hydrostatic pressure increases [Fig. 5(I)]. Glomerular filtration rate [Fig. 5(J)] (in qualitative accordance with Reinhart *et al.*[38]) first decreases and then go to a level slightly higher than the control value. On the other hand, after rsna has been increased the concentrations of renin [Fig. 5(C)] (in qualitative accordance with in Reinhart *et al.*[38] and Braith *et al.*[4]), angiotensin [Fig. 5(D)] (in qualitative accordance with in Braith *et al.*[4]) first make a peak, but then drop to a level higher then the control levels.

Interpretation of the results presented in Fig. 5 requires some further discussion, because they have been obtained by increasing rsna and keeping it at this higher level in spite of the reflex mechanisms, which try to decrease it (rsna inhibition mechanisms). Thus it would be more appropriate to interpret the behavior observed in Fig. 5 as a response to rsna increase combined with impairment of rsna inhibition. This combination is typically encountered in hypertension, congestive heart failure, nephrotic syndrome and cirrhosis.[12] Of course, these diseases do not have the same aetiology. Furthermore, since the proposed model is constructed by combining normal physiological functions, it cannot simulate clinical conditions. Nevertheless, since it includes mechanisms relating rsna, sodium excretion etc, it can provide insight to some causal relations between some common findings (particularly those related to rsna) in these diseases.

The proposed model shows that an increase in rsna causes sodium reabsorption and renin secretion to increase, which is followed by an increase in extracellular fluid volume and consequently increase in blood pressure. This causal relationship between rsna increase and blood

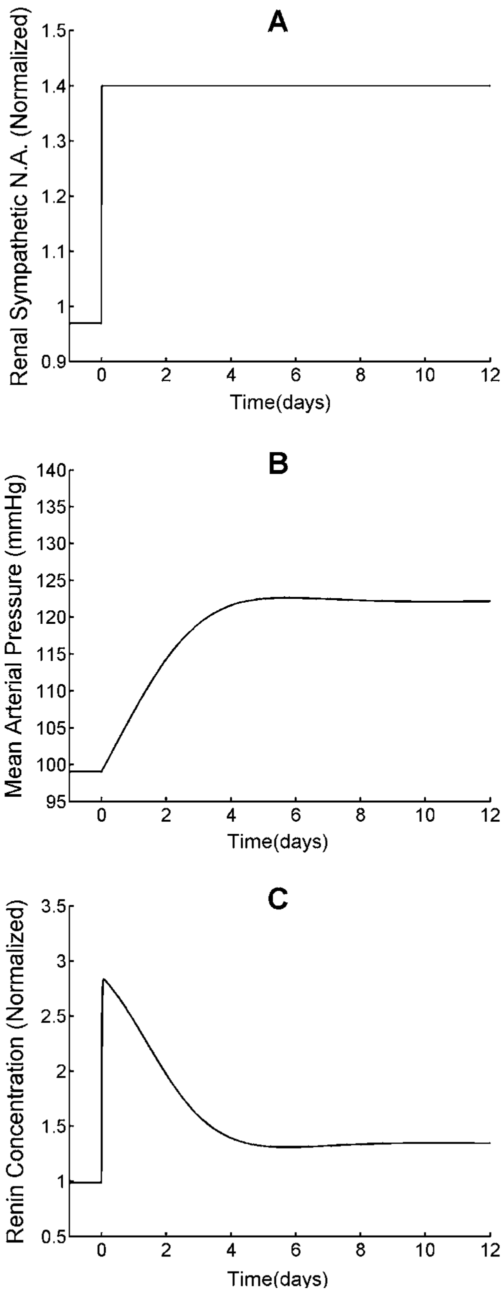

FIGURE 5. Simulation results for rsna increase.

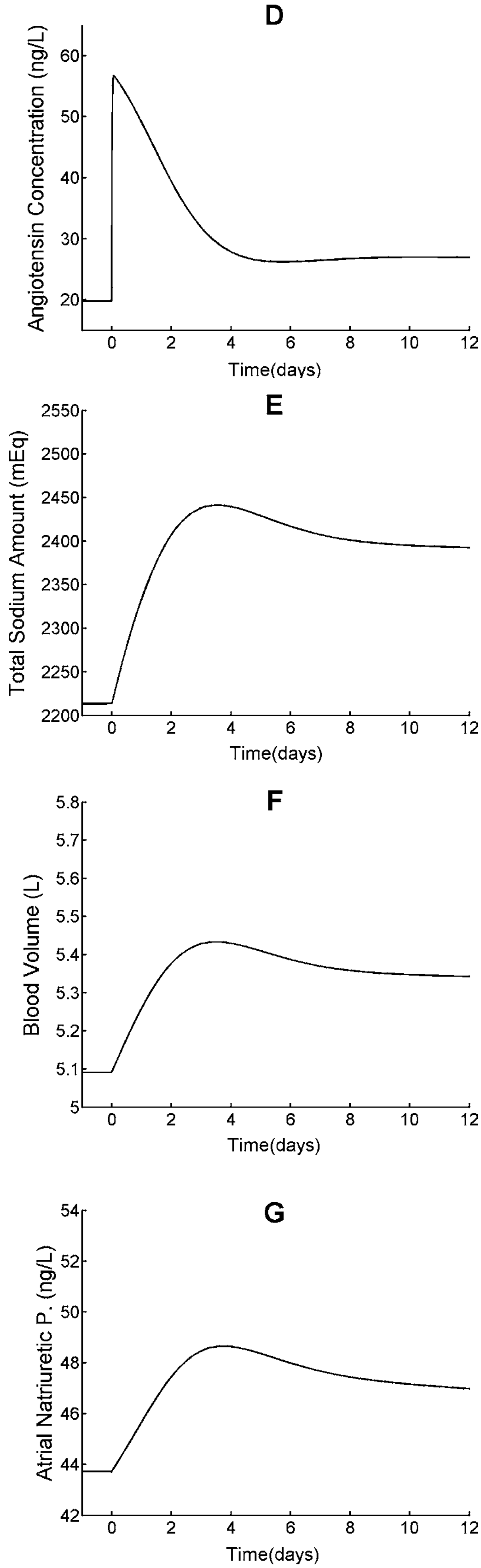

FIGURE 5. Continued.

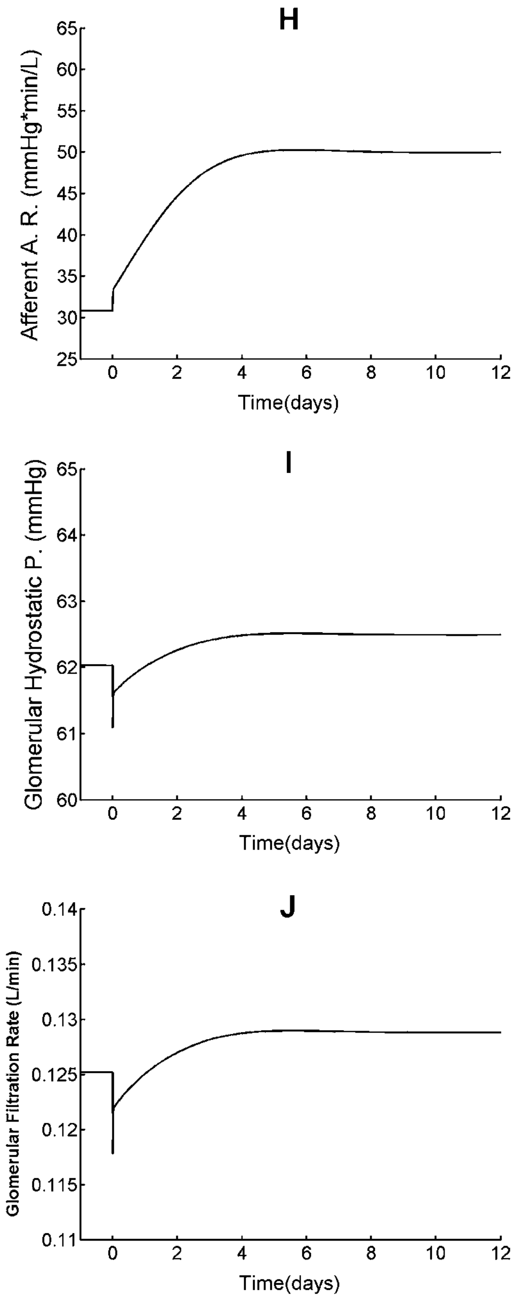

FIGURE 5. Continued.

pressure increase seems to explain the typical hypertension features of high rsna and high blood pressure (rsna increase can be considered as one of the factors contributing to high blood pressure in hypertension disease).

In congestive heart failure the blood pressure is low because the heart fails to pump enough blood. Nevertheless, since the amount of blood entering the kidneys is low, there is a decrease in excretion of water and sodium. Furthermore, renin secretion at the kidneys increases causing higher angiotensin and aldosterone secretion and consequently higher sodium and water reabsorption. As a result, water and sodium retension in the body increases. Additionally, in congestive heart failure rsna is high. When rsna increases, renin secretion and tubular sodium reabsorption increases. In view of the causal relation between high rsna [Fig. 5(A)] and high sodium amount [Fig. 5(E)], the increased rsna in congestive heart failure can be considered as one of the factors contributing to sodium retension.

Also in nephrotic syndrome and cirrhosis high rsna and high sodium amount are observed.[12–14] Our simulation results (Fig. 5) account for a causal relationship between these phenomena.

Hence the proposed model can be said to provide some understanding of the rsna-related mechanisms governing mean arterial pressure increase in hypertension and total sodium increase in congestive heart failure, nephrotic syndrome, and cirrhosis.

Behavior in Case of Renal Denervation

Under excessive sodium intake, blood volume increases causing an increase in arterial pressure and cardiac atrial pressure. Sensed by baroreceptors at cardiac and arterial vessel walls, these increased pressures give rise to a reflex decrease in rsna. This mechanism is referred to as "rsna inhibition under sodium loading." It causes a decrease in renin secretion and tubular sodium reabsorption, hence tends to restore the sodium balance by increasing the sodium excretion. When the rsna inhibition mechanism is somewhat impaired the restoration of sodium balance after sodium loading becomes more difficult.

One way of checking how well the proposed model accounts for "rsna inhibition under sodium loading," is to prevent the inhibition and observe the results. While in simulations this can be achieved by keeping rsna at its steady-state value (preventing the rsna inhibition mechanism under sodium loading), in experiments such a situation can be induced by cutting the renal nerves. However, the two situations are not exactly the same because denervation is a serious pathological situation, where the extra adaptation mechanisms triggered are not included in this model. The resemblance of the simulated situation to the experimental case lies in the fact that rsna cannot respond to sodium loading in both cases.

In the simulation rsna has been kept at its normal value [Fig. 6(A)] while increasing the sodium intake [Fig. 6(B)] 10 times for 10 days. Fig. 6.(C)–(I) present the simulation results with and without rsna blockage. They exhibit qualitative resemblance to experimental results obtained from animals with renal denervation and similar sodium loading,[17,28] in the sense that the mean arterial pressure increase under sodium loading is slightly higher ($\sim$6 mmHg) in the case of rsna blockage as compared to the intact case [Fig. 6(D)]. The increase in mean arterial pressure for compensating rsna blockage is expected to be small because small changes in mean arterial pressure can induce large changes in sodium excretion rate.[18] Also the decrease in renin concentration as a response to sodium loading is slightly weaker for blocked rsna compared to the intact case [Fig. 6(F)], similar to the results given by Greenberg *et al.*[17] The increase in total amount of sodium as a response to sodium loading is slightly stronger for blocked rsna compared to the intact case [Fig. 6(E)], similar to the results in Greenberg *et al.*[17] The increase in artial natriuretic peptide concentration as a response to sodium loading is slightly stronger for blocked rsna compared to the intact

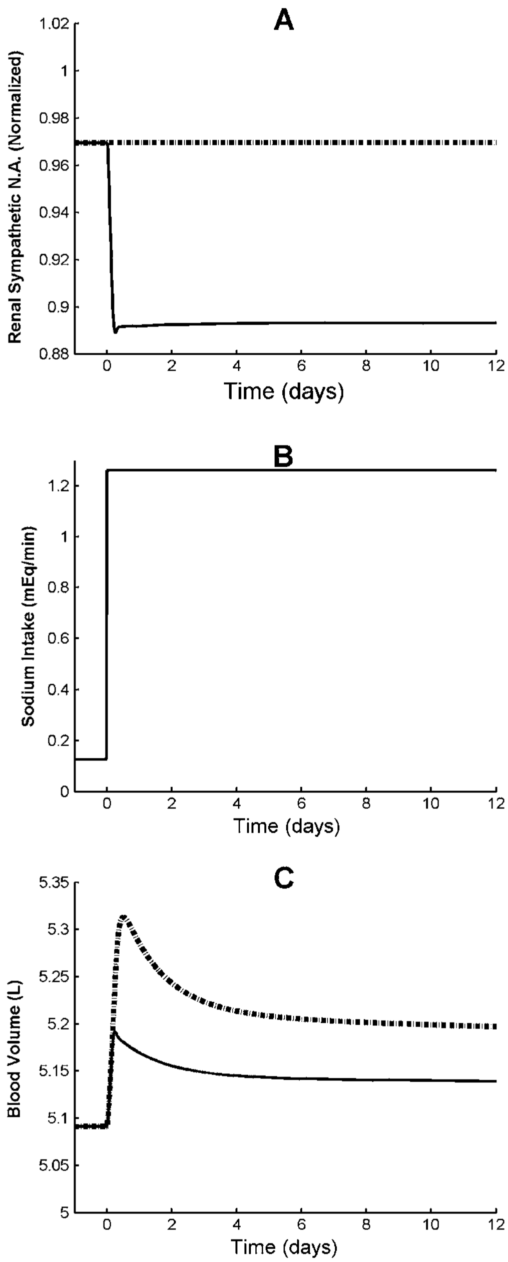

FIGURE 6. Simulation results under sodium loading (10 times increase) with intact (*solid lines*) and blocked (*dotted lines*) renal nerves.

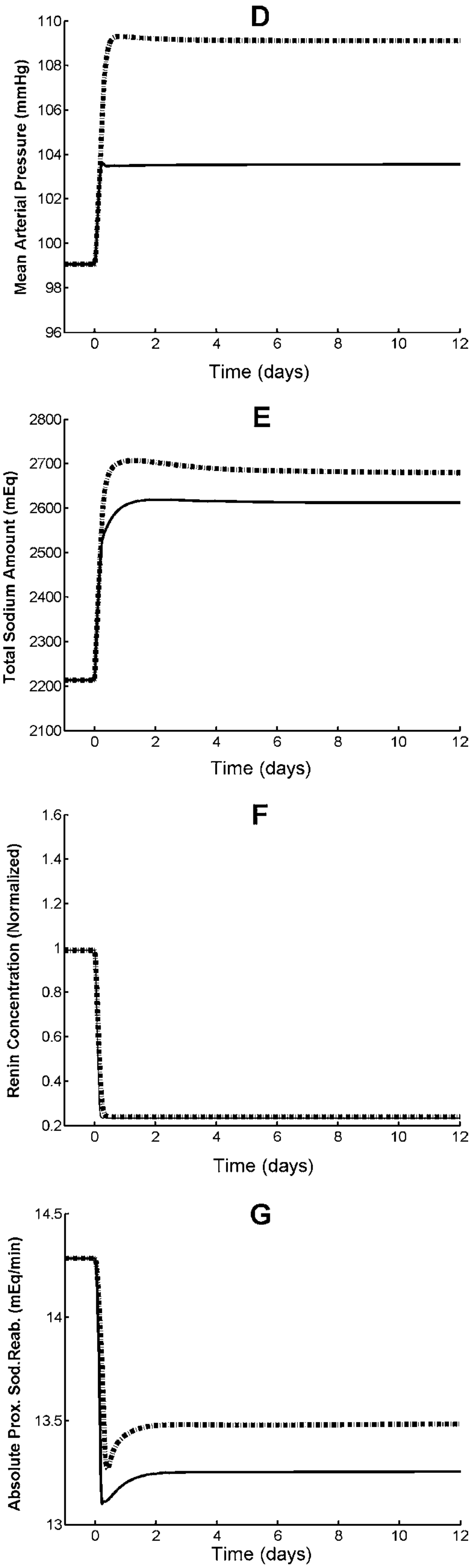

FIGURE 6. Continued.

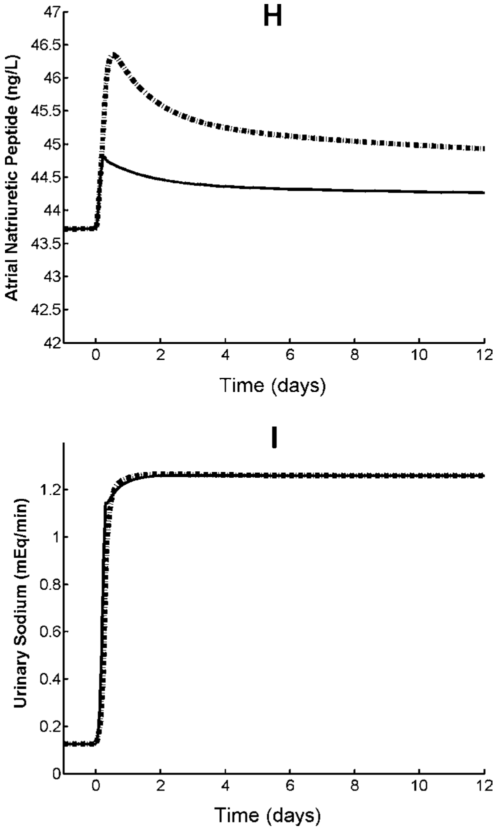

FIGURE 6. Continued.

case [Fig. 6(H)], as theoretically expected (for blocked rsna the sodium accumulation in the body is higher than for the intact case, hence more artial natriuretic peptide is secreted in order to provide higher sodium excretion from the kidneys). The increase in sodium excretion rate is almost the same both for blocked and intact rsna [Fig. 6(I)] (in accordance with the results presented in Greenberg *et al.*[17] and Jacob *et al.*[28]). This can be explained in terms of a compensation induced by the higher increase of the mean arterial pressure and the atrial natriuretic peptide for the blocked rsna. The possibility that under sodium loading the effects of arterial pressure and hormones can mask the influence of rsna on sodium excretion rate has already been mentioned by Lohmeier *et al.*[32]

In Fig. 6(G) the proximal tubular sodium reabsorption rate for the blocked rsna case is higher than for the intact case. This is theoretically expected since increase in rsna directly results in increase in tubular sodium reabsorption. During sodium loading, rsna decreases and causes a decrease in tubular sodium reabsorption. However, in case the rsna is blocked during sodium loading, tubular sodium reabsorption will be higher than intact case.

In Fig. 6(A) with the intact renal nerve (*solid line*) rsna exhibits a decreasing behavior as a response to sodium loading and then remains almost at the same low level because it does not adapt (no resetting). It can be hypothesised that if rsna had an adaptation property, after a while the solid line would have approached the level of the dotted line. Consequently, one would expect also the solid lines in Fig. 6(C), (D), and (E) to approach the dotted lines in the long run. Thus, it can be said that rsna adaptation under sodium loading would have resulted in higher long-term values in blood volume, mean arterial pressure and total sodium amount compared to lack of adaptation. These results support the interpretation of rsna inhibition under sodium loading (volume loading) as a compensatory mechanism for the regulation of body fluid volume. Furthermore, they seem to demonstrate the importance of the lack of rsna resetting in maintaining this compensatory mechanism in the long run.

Simulation results shows that impairment of rsna inhibition under volume loading causes an increase in blood volume and blood pressure. It is known that there is impairment of rsna inhibition to volume loading in salt sensitive hypertension.[12] In view of the causal relation between preventing rsna inhibition to volume loading and high blood pressure, the impairment of the rsna inhibition to volume loading can be considered as one of the factors contributing to high blood pressure in salt sensitive hypertension.

Furthermore; the simulation results showing that the impairment of rsna inhibition under volume loading causes an increase in total sodium amount and it also provides an understanding of a causal relationship between the typical findings (increase of total sodium amount and impairment of rsna inhibition under sodium loading) common to congestive heart failure, nephrotic syndrome, and cirrhosis.[12–14]

CONCLUSION

High basal renal sympathetic nerve activity is known to contribute the pathogenesis of hypertension, congestive heart failure, nephrotic syndrome, and hepatic cirrhosis. In hypertension, congestive heart failure, nephrotic syndrome, and cirrhosis (sodium retention) basal renal sympathetic nerve activity (rsna) is known to be higher than normal. It has been shown that this increased rsna contributes to the pathogenesis of these diseases [Dibona, 1997].[12–14]

In spite of its clinical importance, long-term mathematical models of the cardiovascular system existing in the literature do not sufficiently account for the effect of rsna on kidney functions. Among the mathematical models considered only Guyton's model[22] includes one aspect of renal sympathetic nerve activity, namely the renal vascular resistance.

The objective of the study presented in this paper is the development of a cardiovascular system model that allows analysis of long-term interactions between rsna, arterial pressure, and sodium excretion. The cardiovascular system models developed by Guyton[20,22] and the kidney models developed by Uttamsingh[40] and Coleman[7] have been integrated, while introducing additional mechanisms of direct rsna effects on tubular sodium reabsorption and renin secretion supported by experimental data from literature.[8,35] Furthermore, the dependence of rsna on right atrial pressure and mean arterial pressure,[10,11] as well as the interactions between right atrial pressure, atrial natriuretic peptide, and sodium excretion have been included in the model using experimental open-loop data from literature.[25,33] The resulting mathematical model constitutes the first long-term model of the cardiovascular system accounting for the effects of the rsna on kidney functions in such detail.

Long term tests of the model have been performed under blood volume increase and decrease by altering sodium intake. The results have shown that the model sufficiently well predicts the basic blood volume control mechanisms.

Further tests have been performed by increasing rsna during normal sodium intake, and by blocking rsna during long-term sodium loading. These tests have also resulted in dynamics similar to various experimental and clinical observations.

Simulation results of the proposed model provide an understanding of some the rsna-related mechanisms in various diseases. Among these one can mention the rsna-increase-related increase of mean arterial pressure in hypertension, the rsna-increase-related increase of total sodium amount in congestive heart failure, nephrotic syndrome, and in cirrhosis. Similarly, the model also accounts for mechanisms which cause an increase in mean arterial pressure in salt-sensitive hypertension and an increase in total sodium amount in congestive heart failure, nephrotic syndrome, and cirrhosis related to the impairment of rsna inhibition under sodium loading.

In addition to this, simulation results show that rsna inhibition to volume loading is a compensatory mechanism for the regulation of body fluid volume. And it seems that the lack of rsna resetting is important to maintain this compensatory mechanism in long run.

The model developed is restricted to the regulation mechanisms accounted in Guyton's cardiovascular system models[20,22] and in the kidney models of Coleman[7] and Uttamsingh.[40] Further quantitative improvement can be achieved by including the effect of rsna on the tubular water reabsorption and the effect of aldosterone on sodium reabsorption at the collecting duct.

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